
MTL122

Complex Analysis

Comprehensive Course Notes

A complete, exam-ready treatment following the lecture course:

from the field $\mathbb{C}$ and the topology of the plane, through the Cauchy–Riemann equations, power series and the elementary functions, Cauchy’s theorem and integral formula, the great structural theorems (Liouville, the Fundamental Theorem of Algebra, the Identity, Maximum-Modulus and Open-Mapping theorems), up to isolated singularities, Laurent series and the residue.

How to use these notes

Every definition, theorem, lemma, corollary and proof from the course is included, in the same order and the same notation used in lectures. Worked **examples are boxed in red** so they stand out for revision; key formulas are boxed in amber. Points marked **“NOT true in $\mathbb{R}$ ”** flag exactly where complex analysis parts ways with real analysis — a favourite source of exam questions.

Prepared for end-semester preparation

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CHAPTER 1

The Complex Numbers and the Field $\mathbb{C}$

1.1 The set $\mathbb{C}$ and its identification with $\mathbb{R}^2$

We begin with the definition of the complex numbers.

$$\mathbb{C} = \{ a + ib : a, b \in \mathbb{R}, i^2 = -1 \}.$$

The crucial point is that this set is naturally identified with the plane $\mathbb{R}^2$:

$$\mathbb{R}^2 = \{ (a, b) : a, b \in \mathbb{R} \}, \quad a + ib \longleftrightarrow (a, b).$$

On $\mathbb{R}^2$ we already have componentwise *addition*. Complex multiplication is the extra structure that turns the plane into a field. Written out on ordered pairs:

$$(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2),$$

$$(a_1, b_1) \cdot (a_2, b_2) = (a_1 a_2 - b_1 b_2, a_1 b_2 + a_2 b_1).$$

(The multiplication rule is exactly what one obtains by expanding $(a_1 + ib_1)(a_2 + ib_2)$ and using $i^2 = -1$.)

1.1.1 The real line sits inside the plane

There is an embedding $\mathbb{R} \hookrightarrow \mathbb{R}^2$ given by

$$a \longmapsto (a, 0).$$

Under this identification the real number 1 becomes $(1, 0)$, which is the multiplicative identity, and the arithmetic of the reals is respected. Note the two computations that pin down the role of the second coordinate:

$$(0, 1) \cdot (1, 0) = (0, 1), \quad (0, 1) \cdot (0, 1) = (-1, 0).$$

The second says $i^2 = -1$: the element $i = (0, 1)$ squares to $-1 = (-1, 0)$.

1.2 Multiplication as a geometric operation

1.2.1 Multiplication by a real number

For $r \in \mathbb{R}$,

$$r(a, b) = (ra, rb).$$

Suppose first $r > 0$. Then multiplication by r is a *scaling* (a dilation) of the plane by the factor r ; the vector (a, b) is stretched along its own direction to (ra, rb) , and one may break this single multiplication into two successive multiplications of equal size $\sqrt{r}$.

! Caution

What about $r < 0$? Take $r = -1$. Then $-1 \cdot (a, b) = (-a, -b)$, which is not a mere stretch: geometrically it is a *rotation by 180°* . Thus scaling by a negative real already forces us to think of multiplication as combining a stretch with a rotation.

1.2.2 Multiplication by i is rotation by 90°

Consider the effect of $i = (0, 1)$ on the basis vector $1 = (1, 0)$:

$$1 = (1, 0) \xrightarrow{\times 90^\circ} (0, 1) = i.$$

More generally, from the multiplication rule,

$$i(a, b) = (-b, a),$$

which is precisely the anticlockwise rotation of the vector (a, b) by 90° . *Multiplication by i is a rotation by 90°* . Iterating, $1 \rightarrow i \rightarrow -1 \rightarrow -i \rightarrow 1$ walks around the four quarter-turns of the plane.

Combining a real scaling with a rotation, for $r \in \mathbb{R}$ we also record

$$i r(a, b) = (-rb, ra).$$

Finally, a general complex number $p + iq$ acts on (a, b) by

$$(p + iq)(a, b) = p(a, b) + iq(a, b) = (pa, pb) + (-qb, qa) = (pa - qb, pb + qa),$$

which is exactly the product rule again, and matches the familiar expansion

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc).$$

1.3 $\mathbb{C}$ is a field

$(\mathbb{C}, +, \cdot)$ is a **field**.

Addition and multiplication are commutative, associative and distributive; $(0, 0)$ is the additive identity, $(1, 0)$ the multiplicative identity, and every nonzero $a + ib$ has the inverse

$$(a + ib)^{-1} = \frac{a - ib}{a^2 + b^2}.$$

1.4 Matrix representation of complex numbers

Complex multiplication can be encoded by real 2×2 matrices. Fix $a + ib$ and consider the map “multiply by (a, b) ”:

$$(c, d) \xrightarrow{(a, b) \cdot} (ac - bd, ad + bc).$$

As a linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ acting on the column $\begin{pmatrix} c \\ d \end{pmatrix}$,

$$\begin{pmatrix} c \\ d \end{pmatrix} \mapsto \begin{pmatrix} ac - bd \\ ad + bc \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix}.$$

This gives the identification

$$a + ib = (a, b) \longleftrightarrow \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

One checks directly that matrix multiplication reproduces complex multiplication:

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix},$$

which is the matrix of $(a, b) \cdot (c, d) = (ac - bd, ad + bc)$. Hence

$$\mathbb{C} \longleftrightarrow \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

1.5 Polar representation

A complex number corresponds to a vector in $\mathbb{R}^2$, and a vector is described by a *direction* and a *modulus*. Given $a + ib$, write $r = \sqrt{a^2 + b^2}$ for its length and seek an angle θ with

$$a = r \cos \theta, \quad b = r \sin \theta.$$

Then

$$a + ib = r \cos \theta + i r \sin \theta = r(\cos \theta + i \sin \theta) = r e^{i\theta},$$

the **polar representation**, where

$$r = \sqrt{a^2 + b^2}, \quad \tan \theta = \frac{b}{a}, \quad r^2 = a^2 + b^2.$$

Definition 1.1: Principal argument

We ask for a value $\theta \in (-\pi, \pi]$ satisfying the above. Such a value of θ is called the **principal argument** of $a + ib$.

Remark

The choice of θ is *not* unique: θ and $\theta + 2\pi n$ ($n \in \mathbb{Z}$) give the same point. Fixing $\theta \in (-\pi, \pi]$ removes the ambiguity and defines the principal argument.

1.5.1 Multiplication in polar form

Polar form makes multiplication transparent. If

$$a + ib = r_1 e^{i\theta_1}, \quad c + id = r_2 e^{i\theta_2},$$

then

$$(a + ib)(c + id) = r_1 r_2 e^{i(\theta_1 + \theta_2)} \pmod{2\pi}.$$

Moduli multiply and arguments add (modulo 2π). This is the analytic form of the “stretch-and-rotate” picture: the factor r_2 scales, and the angle θ_2 rotates.

1.6 Modulus and conjugate

The number r is called the **modulus** of the complex number $z = a + ib$, written $|z| = r$. The **conjugate** of $z = a + ib$ is

$$\bar{z} = a - ib.$$

The fundamental relations are

$$r^2 = z \bar{z}, \quad \text{i.e.} \quad |z|^2 = z \bar{z}.$$

Indeed $z \bar{z} = (a + ib)(a - ib) = a^2 + b^2$.

CHAPTER 2

The Extended Complex Plane and Stereographic Projection

2.1 Complex numbers as points on a sphere

We wish to set up a correspondence

complex numbers $\longleftrightarrow$ points on a sphere.

Take the sphere (the **Riemann sphere**)

$$S = \left\{ (u, v, w) : u^2 + v^2 + \left(w - \frac{1}{2}\right)^2 = \frac{1}{4} \right\},$$

which sits on the xy -plane, has its south pole at the origin $(0, 0, 0)$ and its north pole at $(0, 0, 1)$.

2.1.1 The projection

Take a point (u, v, w) on S . Draw the line joining the north pole $(0, 0, 1)$ and the point (u, v, w) . Suppose it cuts the xy -plane at the point $(x, y, 0)$. Then we **identify** the sphere point (u, v, w) with the complex number $x + iy$.

The three points $(0, 0, 1)$, (u, v, w) and $(x, y, 0)$ are collinear, so

$$\frac{u - 0}{x - 0} = \frac{v - 0}{y - 0} = \frac{w - 1}{-1}.$$

Solving the first two equalities against the third gives the projection formulas

$$\boxed{x = \frac{u}{1 - w}, \quad y = \frac{v}{1 - w}.}$$

2.2 The inverse map and its injectivity

Example 2.1: Recovering (u, v, w) from (x, y)

For a given (x, y) we recover (u, v, w) by

$$u = \frac{x}{x^2 + y^2 + 1}, \quad v = \frac{y}{x^2 + y^2 + 1}, \quad w = \frac{x^2 + y^2}{x^2 + y^2 + 1}.$$

Derivation. From $x = \frac{u}{1 - w}$ and $y = \frac{v}{1 - w}$ we get $\frac{x}{y} = \frac{u}{v}$, so $v = \frac{uy}{x}$. Also $1 - w = \frac{u}{x}$, hence $\frac{1}{1 - w} = \frac{x}{u}$. Substituting into the sphere equation $u^2 + v^2 + \left(w - \frac{1}{2}\right)^2 = \frac{1}{4}$, equivalently

$u^2 + v^2 = w - w^2 = w(1 - w)$, and simplifying yields

$$u^2 \left(1 + \frac{y^2}{x^2} + \frac{1}{x^2} \right) = \frac{w}{x} \cdot \frac{u}{x} \cdot \frac{1}{\dots},$$

which reduces to $u = \frac{x}{x^2 + y^2 + 1}$, and the expressions for v, w follow.

2.2.1 Injectivity

We check the injectivity of the correspondence. Suppose two sphere points (u_1, v_1, w_1) and (u_2, v_2, w_2) project to the same plane point, so

$$\frac{u_1}{1 - w_1} = \frac{u_2}{1 - w_2}, \quad \frac{v_1}{1 - w_1} = \frac{v_2}{1 - w_2}.$$

Then

$$\frac{u_1}{u_2} = \frac{v_1}{v_2} = \frac{1 - w_1}{1 - w_2} = \lambda \quad (\text{say}).$$

So $1 - w_1 = \lambda(1 - w_2)$, i.e. $w_2 - w_1 = \lambda - \lambda w_2 - \frac{1}{2} + \frac{1}{2}$. Using $u_i^2 + v_i^2 + (w_i - \frac{1}{2})^2 = \frac{1}{4}$ for $i = 1, 2$ and $u_1 = \lambda u_2, v_1 = \lambda v_2$, one computes

$$\lambda^2 u_2^2 + \lambda^2 v_2^2 + \left(\lambda - \lambda w_2 - \frac{1}{2} \right)^2 = \frac{1}{4},$$

and after simplification (expanding and using $u_2^2 + v_2^2 = w_2(1 - w_2)$)

$$\lambda^2 - \lambda^2 w_2 - \lambda(1 - w_2) = 0 \implies \lambda(\lambda - 1)(1 - w_2) = 0.$$

Since $w_2 \neq 1$, either $\lambda = 0$ (which forces $w_1 = 1$, impossible) or $\lambda = 1$. Hence $\lambda = 1$ and

$$(u_1, v_1, w_1) = (u_2, v_2, w_2).$$

The correspondence is injective.

2.3 The point at infinity

Compute $x^2 + y^2$ in terms of w . Using $u^2 + v^2 = w(1 - w)$,

$$x^2 + y^2 = \frac{u^2}{(1 - w)^2} + \frac{v^2}{(1 - w)^2} = \frac{u^2 + v^2}{(1 - w)^2} = \frac{w(1 - w)}{(1 - w)^2} = \frac{w}{1 - w}.$$

Consequently:

- If $x^2 + y^2 \rightarrow \infty$, then $w \rightarrow 1$.
- If $w \rightarrow 1$, then $u^2 + v^2 \rightarrow 0$, i.e. $u, v \rightarrow 0$.

So, if $|z|^2 = x^2 + y^2 \rightarrow \infty$, then $(u, v, w) \rightarrow (0, 0, 1)$. The north pole $(0, 0, 1)$ is therefore called the **point at infinity** in the complex plane; adding it to $\mathbb{C}$ gives the extended complex plane $\mathbb{C} \cup \{\infty\}$, in bijection with the whole sphere S .

Example 2.2: Image of a spherical cap is the exterior of a disc

Let $U = \{(u, v, w) \in S : w \geq w_0\}$ for a fixed $0 < w_0 < 1$, and let $V \subseteq \mathbb{C}$ be the corresponding set in the plane. Writing $z = x + iy = \frac{u}{1-w} + i\frac{v}{1-w}$, we found $|z|^2 = \frac{w}{1-w}$. On a unit sphere-type computation this rearranges to

$$w = \frac{|z|^2}{|z|^2 + 1}.$$

The condition $w \geq w_0$ becomes $\frac{|z|^2}{|z|^2 + 1} \geq w_0$, i.e. $|z|^2 \geq \frac{w_0}{1-w_0}$, that is

$$|z| \geq \sqrt{\frac{w_0}{1-w_0}} =: R.$$

Thus the image is

$$V = \{z \in \mathbb{C} : |z| \geq R\}, \quad R = \sqrt{\frac{w_0}{1-w_0}},$$

the *exterior* of the circle centred at the origin with radius R . Geometrically, a cap of the sphere containing the north pole maps to the outside of a disc, consistent with the north pole being the point at infinity.

CHAPTER 3

Topology of the Complex Plane

3.1 The metric on $\mathbb{C}$

$\mathbb{C}$ becomes a metric space through the absolute value. For $z = a + ib$,

$$|z|^2 = z\bar{z} = a^2 + b^2, \quad |z| = \sqrt{a^2 + b^2}.$$

Define $d : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{R}$ by

$$d(z_1, z_2) = |z_1 - z_2| = |(a_1 - a_2) + i(b_1 - b_2)| = \sqrt{(a_1 - a_2)^2 + (b_1 - b_2)^2}.$$

This is exactly the Euclidean (ℓ_2) distance on $\mathbb{R}^2$, so

$$(\mathbb{C}, |\cdot|) \cong (\mathbb{R}^2, \ell_2)$$

as metric spaces. The **open ball** of radius r about z is

$$B(z, r) = \{w \in \mathbb{C} : d(z, w) < r\} = \{w : |z - w| < r\}.$$

3.2 Convergence of sequences

Definition 3.1: Convergence

We say $z_n \rightarrow z$ if, given any $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that

$$|z_n - z| < \varepsilon \quad \text{for all } n \geq n_0.$$

The link with the real and imaginary parts rests on a simple inequality. For any complex number z ,

$$|\operatorname{Im}(z)|, |\operatorname{Re}(z)| \leq |z| \leq |\operatorname{Im}(z)| + |\operatorname{Re}(z)|.$$

Applying this to $z_n - z$: from $|z_n - z| < \varepsilon$ we get $|\operatorname{Re}(z_n - z)|, |\operatorname{Im}(z_n - z)| < \varepsilon$. Hence

$$z_n \rightarrow z \implies \operatorname{Re}(z_n) \rightarrow \operatorname{Re}(z), \quad \operatorname{Im}(z_n) \rightarrow \operatorname{Im}(z),$$

and conversely, so that

$$a_n \rightarrow a, \quad b_n \rightarrow b \implies a_n + ib_n \rightarrow a + ib.$$

Convergence in $\mathbb{C}$ is exactly coordinatewise convergence.

3.3 The topology of $\mathbb{C}$

The topology on $\mathbb{C}$ is nothing but the product topology on $\mathbb{R}^2$ coming from $\mathbb{R}$. We record the basic facts.

- $\mathbb{C}$ is a **complete** metric space (every Cauchy sequence converges).
- **Heine–Borel:** the compact subsets of $\mathbb{C}$ are exactly those which are *closed and bounded*.
- **Connectedness:** a set that is *open and connected* is *path connected*, and in fact *polygonally* (“stair”) *connected* — any two points can be joined by a path made of horizontal and vertical segments lying in the set.

Remark

The stair-connectedness of open connected sets is used repeatedly: to prove a holomorphic function with zero derivative is constant, one moves along horizontal and vertical segments and applies the one-variable mean value theorem on each.

3.4 Continuous functions

Definition 3.2: Continuity

Let $U \subseteq \mathbb{C}$ and let $f : U \rightarrow \mathbb{C}$ be given. For $z \in U$, we say f is **continuous at z** if for any sequence $z_n \rightarrow z$ we have $f(z_n) \rightarrow f(z)$.

Every complex function splits into real and imaginary parts. Writing $f(a + ib) = c + id$,

$$\operatorname{Re}(f)(a + ib) = c, \quad \operatorname{Im}(f)(a + ib) = d, \quad f = \operatorname{Re}(f) + i \operatorname{Im}(f),$$

with $\operatorname{Re}(f), \operatorname{Im}(f) : U \rightarrow \mathbb{R}$ real-valued functions of (a, b) .

Theorem 3.1: Continuity via real and imaginary parts

Suppose $z_n \rightarrow z$. Then $f(z_n) \rightarrow f(z)$ if and only if

$$\operatorname{Re}(f)(z_n) \rightarrow \operatorname{Re}(f)(z) \quad \text{and} \quad \operatorname{Im}(f)(z_n) \rightarrow \operatorname{Im}(f)(z).$$

Consequently, f is continuous at z if and only if $\operatorname{Re}(f)$ and $\operatorname{Im}(f)$ are continuous at z .

Proof. Immediate from the inequality $|\operatorname{Re}(\zeta)|, |\operatorname{Im}(\zeta)| \leq |\zeta| \leq |\operatorname{Re}(\zeta)| + |\operatorname{Im}(\zeta)|$ applied to $\zeta = f(z_n) - f(z)$: the left inequality gives the forward implication, the right inequality the reverse. ■

CHAPTER 4

Differentiability

Before defining complex differentiability we recall the real theory in the three cases $\mathbb{R} \rightarrow \mathbb{R}$, $\mathbb{R}^2 \rightarrow \mathbb{R}$ and $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, since complex differentiability will be a strengthening of real ($\mathbb{R}^2 \rightarrow \mathbb{R}^2$) differentiability.

4.1 Real differentiability: one variable

Definition 4.1: Differentiability, $f : U(\subseteq \mathbb{R}) \rightarrow \mathbb{R}$

Let $f : U(\subseteq \mathbb{R}) \rightarrow \mathbb{R}$ be given. We say f is **differentiable** at $a \in U$ if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists in } \mathbb{R}.$$

This limit is denoted $f'(a)$.

4.1.1 Equivalent formulation

In a neighbourhood of a we may write

$$f(a+h) - f(a) = \alpha h + h \varepsilon(h),$$

where α is a constant and $\varepsilon(h)$ is a function which goes to 0 as $h \rightarrow 0$. (Here $\alpha = f'(a)$.) This “linear approximation” viewpoint is the one that generalises.

4.2 Real differentiability: $f : U(\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$

Definition 4.2: Differentiability, $f : U(\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$

We say f is **differentiable** at a point $(a, b) \in U$ if we can write, in a neighbourhood of (a, b) ,

$$f(a+h, b+k) - f(a, b) = \alpha h + \beta k + h \varepsilon_1(h, k) + k \varepsilon_2(h, k),$$

where α, β are constants and $\varepsilon(h, k) \rightarrow 0$ as $h, k \rightarrow 0$.

4.3 Real differentiability: $f : U(\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}^2$

Definition 4.3: Differentiability, $f : U(\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}^2$

We say f is **differentiable** at $(a, b) \in U$ if there exists a linear transformation $T_{(a,b)} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that

$$f(a+h, b+k) - f(a, b) = T_{(a,b)}(h, k) + \|(h, k)\| E_{(a,b)}(h, k),$$

where $E_{(a,b)}(h, k) \rightarrow (0, 0)$ as $\|(h, k)\| \rightarrow 0$, and $\|(h, k)\| = \sqrt{h^2 + k^2}$. Equivalently,

$$\lim_{(h,k) \rightarrow (0,0)} \frac{\|f(a+h, b+k) - f(a, b) - T_{(a,b)}(h, k)\|}{\|(h, k)\|} = 0.$$

4.3.1 Linear transformations are differentiable

$T_{(a,b)}$ is a linear transformation; the matrix of this linear transformation, with respect to the standard ordered basis, is the Jacobian

$$\begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix}, \quad \text{where } f = (f_1, f_2).$$

Example 4.1: Two real-differentiable maps

- (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(a, b) = (a, -b)$, is a linear transformation; hence it is differentiable.
- (b) $\gamma : \mathbb{C} \rightarrow \mathbb{C}$, $z \mapsto \bar{z}$, i.e. $(a, b) \mapsto (a, -b)$, is $\mathbb{R}$ -differentiable. (Here $\mathbb{C}$ is viewed as the base field $\mathbb{R}^2$.) We will see below that although it is real-differentiable, it is *not* complex differentiable.

4.4 Complex differentiability

Definition 4.4: Complex differentiability

Let $f : U(\subseteq \mathbb{C}) \rightarrow \mathbb{C}$ be given and $a \in U$. We say f is **(complex) differentiable** at a if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad \text{exists in } \mathbb{C}.$$

We denote the limit by $f'(a)$. *Crucially, here h is complex* — the limit must exist as $h \rightarrow 0$ through *all* complex directions, not just real ones.

Rewriting, $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} - f'(a) = 0$ gives

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - f'(a)h}{h} = 0 \implies \lim_{h \rightarrow 0} \frac{|f(a+h) - f(a) - f'(a)h|}{|h|} = 0.$$

4.4.1 Complex differentiable $\Rightarrow$ real differentiable

Comparing the last display with Definition 4.3, a function which is complex differentiable at a point $a \in U$ is also $\mathbb{R}^2$ -differentiable there, when we regard $a = (b, c)$ in $\mathbb{R}^2$; the linear map is $h \mapsto f'(a)h$ (multiplication by the complex number $f'(a)$).

! Caution

What about the converse? Real differentiability does *not* imply complex differentiability. We know that $\gamma : \mathbb{C} \rightarrow \mathbb{C}$, $z \mapsto \bar{z}$, is real differentiable (it is a linear transformation). But

$$\lim_{h \rightarrow 0} \frac{\gamma(a+h) - \gamma(a)}{h} = \lim_{h \rightarrow 0} \frac{\bar{h}}{h} = \begin{cases} 1, & h \in \mathbb{R} \text{ (real axis)}, \\ -1, & h \in i\mathbb{R} \text{ (imaginary axis)}. \end{cases}$$

The two directional limits disagree, so the limit **does not exist**: $z \mapsto \bar{z}$ is not complex differentiable anywhere. Thus

complex differentiable $\implies$ real differentiable, but the converse need not be true.

4.4.2 The matrix of $f'(a)$ and the first appearance of Cauchy–Riemann

If we regard the complex-linear map $h \mapsto f'(a)h$ as a real-linear map and write $f'(a) = c + id$, its matrix is the multiplication matrix

$$[f'(a)] = [c + id] = \begin{pmatrix} c & -d \\ d & c \end{pmatrix}.$$

On the other hand, viewing $f = (f_1, f_2)$ as a map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, the matrix of the derivative with respect to the standard ordered basis is the Jacobian

$$\begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix}.$$

Comparing the two matrices entry by entry forces

$$\frac{\partial f_1}{\partial x} = \frac{\partial f_2}{\partial y}, \quad \frac{\partial f_1}{\partial y} = -\frac{\partial f_2}{\partial x}.$$

These are the **Cauchy–Riemann equations**, developed fully in the next chapter.

Example 4.2: Basic complex-differentiable maps

The following are (complex) differentiable on $\mathbb{C}$:

$$z \mapsto z, \quad z \mapsto z^2, \quad z \mapsto P(z) \text{ (any polynomial)}, \quad z \mapsto e^z.$$

Remark

If f is differentiable at $a \in U$, then f is continuous at a :

$$\lim_{h \rightarrow 0} (f(a+h) - f(a)) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot \lim_{h \rightarrow 0} h = f'(a) \cdot 0 = 0.$$

4.5 Algebra of differentiable functions

Proposition 4.1: Algebra of derivatives

Let $f, g : U \rightarrow \mathbb{C}$ be given, $a \in U$, and suppose f, g are differentiable at a . Then:

(1) $f \pm g$, fg and cf are differentiable at a , with

$$(f \pm g)'(a) = f'(a) \pm g'(a), \quad (fg)'(a) = f'(a)g(a) + f(a)g'(a), \quad (cf)'(a) = c f'(a).$$

(2) If $g(a) \neq 0$, then f/g is differentiable at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - g'(a)f(a)}{g(a)^2}.$$

In particular $\left(f \cdot \frac{1}{g}\right)'(a)$ is given by the same expression.

4.6 The chain rule

Theorem 4.1: Chain rule

Let $g : U \rightarrow \mathbb{C}$ and $f : V \rightarrow \mathbb{C}$ be differentiable with $g(U) \subseteq V$. Then $f \circ g : U \rightarrow \mathbb{C}$ is differentiable and

$$(f \circ g)'(z) = f'(g(z)) g'(z).$$

Corollary 4.1

$$\left(\frac{1}{g(z)}\right)' = -\frac{1}{g(z)^2} g'(z).$$

Proof of the chain rule. Take $z_0 \in U$; we must show

$$\lim_{h \rightarrow 0} \frac{f(g(z_0 + h)) - f(g(z_0))}{h} \text{ exists and equals } f'(g(z_0)) g'(z_0) =: S.$$

Let h_n be a sequence with $h_n \rightarrow 0$. We analyse

$$\frac{f(g(z_0 + h_n)) - f(g(z_0))}{h_n} = \frac{f(g(z_0 + h_n)) - f(g(z_0))}{g(z_0 + h_n) - g(z_0)} \cdot \frac{g(z_0 + h_n) - g(z_0)}{h_n}.$$

Case 1. Suppose for all but finitely many n we have $g(z_0 + h_n) - g(z_0) \neq 0$. Then the first factor $\rightarrow f'(g(z_0))$ (as $h_n \rightarrow 0$ forces $g(z_0 + h_n) \rightarrow g(z_0)$ by continuity of g) and the second factor $\rightarrow g'(z_0)$. Their product $\rightarrow S$.

Case 2. Suppose $g(z_0 + h_n) = g(z_0)$ for infinitely many n . Divide the indices into two (infinite) sets

$$I_1 = \{n : g(z_0 + h_n) = g(z_0)\}, \quad I_2 = \{n : g(z_0 + h_n) \neq g(z_0)\}.$$

Along I_1 , the difference quotient for g is 0, so

$$g'(z_0) = \lim_{\substack{n \rightarrow \infty \\ n \in I_1}} \frac{g(z_0 + h_n) - g(z_0)}{h_n} = 0,$$

and also, since $f(g(z_0 + h_n)) = f(g(z_0))$ for $n \in I_1$,

$$\lim_{\substack{n \rightarrow \infty \\ n \in I_1}} \frac{f(g(z_0 + h_n)) - f(g(z_0))}{h_n} = 0 = S.$$

Along I_2 we are in the situation of Case 1:

$$\lim_{\substack{n \rightarrow \infty \\ n \in I_2}} \frac{f(g(z_0 + h_n)) - f(g(z_0))}{h_n} = f'(g(z_0)) g'(z_0) = S.$$

Both subsequences tend to S , hence the whole sequence does. As $h_n \rightarrow 0$ was arbitrary, the limit exists and equals S . ■

CHAPTER 5

The Cauchy–Riemann Equations

5.1 Necessary condition for differentiability

Theorem 5.1: Cauchy–Riemann (necessity)

Let $f : U \rightarrow \mathbb{C}$ be such that f is differentiable at a point $z_0 \in U$. Then the partial derivatives f_x and f_y at z_0 exist and satisfy the identity

$$f_y(z_0) = i f_x(z_0).$$

Recall the meaning of the partials. With $z_0 = a + ib$ and $h \in \mathbb{R}$,

$$\begin{aligned} f_x(z_0) &= \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} = \lim_{h \rightarrow 0} \frac{f(a + h + ib) - f(a + ib)}{h}, \\ f_y(z_0) &= \lim_{h \rightarrow 0} \frac{f(a + i(b + h)) - f(a + ib)}{h}. \end{aligned}$$

Remark

If f is differentiable at z_0 , then $f'(z_0) = f_x(z_0)$.

Proof. Given that f is differentiable at z_0 , the limit

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists and is *independent of the path along which* $h \rightarrow 0$.

Path along the real axis. Choose $h \rightarrow 0$ with $h \in \mathbb{R}$. Then the difference quotient is exactly the definition of f_x , so

$$f'(z_0) = f_x(z_0).$$

Path along the imaginary axis. We also have $f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$ with $h \in i\mathbb{R}$. Writing $h = ik$ with $k \in \mathbb{R}$ and $k \rightarrow 0$,

$$f'(z_0) = \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{f(a + ib + ik) - f(a + ib)}{ik} = \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{f(a + i(b + k)) - f(a + ib)}{ik} = \frac{1}{i} f_y(z_0).$$

Hence

$$f_x(z_0) = f'(z_0) = \frac{1}{i} f_y(z_0) \implies \boxed{f_y(z_0) = i f_x(z_0)}. \quad \blacksquare$$

5.2 Cauchy–Riemann in terms of u and v

Write $f = u + iv$ with $u, v : U \rightarrow \mathbb{R}$, $u = \operatorname{Re}(f)$, $v = \operatorname{Im}(f)$. Then

$$f_x = u_x + iv_x, \quad f_y = u_y + iv_y.$$

The identity $f_y = if_x$ reads

$$u_y + iv_y = i(u_x + iv_x) = -v_x + iu_x,$$

and equating real and imaginary parts gives the classical form:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad (\text{at } z_0).$$

5.3 The converse fails

! Caution

What about the converse? Let $f : U \rightarrow \mathbb{C}$ satisfy $f_y(z_0) = if_x(z_0)$ at some point $z_0 \in U$. Does it follow that f is differentiable at z_0 ? **No.**

Example 5.1: CR at a point does not imply differentiability

Consider

$$f(z) = f(x, y) = \begin{cases} \frac{xy(x + iy)}{x^2 + y^2}, & z \neq 0, \\ 0, & z = 0, \end{cases} \quad \text{i.e.} \quad f(z) = \begin{cases} \frac{\operatorname{Re}(z) \operatorname{Im}(z) z}{|z|^2}, & z \neq 0, \\ 0, & z = 0. \end{cases}$$

Partials at 0. Along the axes,

$$f_x(0) = \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{f(0 + h) - f(0)}{h} = 0, \quad f_y(0) = \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{f(0 + ih) - f(0)}{h} = 0,$$

so $f_y(0) = if_x(0)$ holds ($0 = i \cdot 0$): the Cauchy–Riemann equations are satisfied at 0.

But the derivative does not exist. Write $h = h_1 + ih_2$. Then

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\operatorname{Re}(h) \operatorname{Im}(h) h}{|h|^2 h} = \lim_{(h_1, h_2) \rightarrow 0} \frac{h_1 h_2}{h_1^2 + h_2^2}.$$

Taking $h_2 = mh_1$ this equals $\frac{m}{1 + m^2}$, which depends on the direction m . The limit **does not exist**, so f is not differentiable at 0 even though the Cauchy–Riemann equations hold there.

5.4 A sufficient condition (partial converse)

The remedy is to add continuity of the partials. First the underlying real-analysis fact.

Theorem 5.2: Result from calculus

Let $f : U(\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$ be such that f_x, f_y exist at $(a, b) \in U$, and one of them exists in a neighbourhood of (a, b) and is continuous at (a, b) . Then f is (real) differentiable at (a, b) .

Theorem 5.3: Partial converse of Cauchy–Riemann

Let $f : U \rightarrow \mathbb{C}$ be given, and suppose f_x, f_y exist in U , with $f_y = if_x$ at z_0 , and f_x, f_y continuous in U . Then f is differentiable at z_0 .

Proof. We must show

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} \text{ exists and equals } u_x(z_0) + iv_x(z_0).$$

Write $z_0 = a_0 + ib_0 = (a_0, b_0)$ and $h = a + ib = (a, b)$, and $f = u + iv$. Consider the increment of u , split by moving first in y then in x :

$$\frac{u(z_0 + h) - u(z_0)}{h} = \frac{u(a_0 + a, b_0 + b) - u(a_0 + a, b_0) + u(a_0 + a, b_0) - u(a_0, b_0)}{a + ib}.$$

Group and insert the factors b and a :

$$= \frac{u(a_0 + a, b_0 + b) - u(a_0 + a, b_0)}{b} \cdot \frac{b}{a + ib} + \frac{a}{a + ib} \cdot \frac{u(a_0 + a, b_0) - u(a_0, b_0)}{a}.$$

Apply the one-variable **Mean Value Theorem** in each bracket (in the first, the x -coordinate is fixed at $a_0 + a$ and we vary y ; in the second we vary x):

$$= \frac{b}{a + ib} u_y(a_0 + a, b_0 + \lambda b) + \frac{a}{a + ib} u_x(a_0 + \mu a, b_0), \quad \lambda, \mu \in (0, 1).$$

Likewise for v ,

$$\frac{v(z_0 + h) - v(z_0)}{h} = \frac{b}{a + ib} v_y(a_0 + a, b_0 + \lambda' b) + \frac{a}{a + ib} v_x(a_0 + \mu' a, b_0), \quad \lambda', \mu' \in (0, 1).$$

Adding, and writing the evaluation points collectively as $z_1, z_2, z_3, z_4 \rightarrow z_0$ as $h \rightarrow 0$,

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{b}{a + ib} [u_y(z_1) + iv_y(z_2)] + \frac{a}{a + ib} [u_x(z_3) + iv_x(z_4)].$$

Compare this with the candidate limit. Using $\frac{a + ib}{a + ib} f_x(z_0)$ and the Cauchy–Riemann identity $f_y(z_0) = if_x(z_0)$,

$$f_x(z_0) = \frac{a + ib}{a + ib} f_x(z_0) = \frac{a}{a + ib} f_x(z_0) + \frac{ib}{a + ib} f_x(z_0) = \frac{a}{a + ib} f_x(z_0) + \frac{b}{a + ib} f_y(z_0),$$

i.e.

$$f_x(z_0) = \frac{a}{a + ib} [u_x(z_0) + iv_x(z_0)] + \frac{b}{a + ib} [u_y(z_0) + iv_y(z_0)].$$

Subtracting,

$$\begin{aligned} \left| \frac{f(z_0 + h) - f(z_0)}{h} - f_x(z_0) \right| &\leq \left| \frac{a}{a + ib} \right| |(u_x(z_3) - u_x(z_0)) + i(v_x(z_4) - v_x(z_0))| \\ &\quad + \left| \frac{b}{a + ib} \right| |(u_y(z_1) - u_y(z_0)) + i(v_y(z_2) - v_y(z_0))|. \end{aligned}$$

Since $\left| \frac{a}{a + ib} \right|, \left| \frac{b}{a + ib} \right| \leq 1$ and the partials are continuous, as $h \rightarrow 0$ (so $z_1, z_2, z_3, z_4 \rightarrow z_0$) the right-hand side $\rightarrow 0$. Therefore

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} - f_x(z_0) = 0,$$

so the derivative exists at z_0 . ■

Example 5.2: Points of differentiability of $f(z) = |z|^2$

Consider $f : \mathbb{C} \rightarrow \mathbb{C}$, $f(z) = |z|^2$, i.e. $f(x, y) = x^2 + y^2$ (so $u = x^2 + y^2$, $v = 0$). **Check Cauchy–Riemann.**

$$f_x = 2x, \quad f_y = 2y, \quad u_x = 2x, \quad u_y = 2y, \quad v_x = v_y = 0.$$

The condition $f_y = if_x$ reads $2y = i2x$, i.e. $2y = 2ix$, which (comparing) forces $x = 0$ and $y = 0$. Hence the Cauchy–Riemann equations are satisfied *only at* $z = 0$. Moreover f_x, f_y are continuous everywhere, in particular in a neighbourhood of 0; by the partial-converse theorem, f is differentiable at 0.

Remark

So $f(z) = |z|^2$ is differentiable *only* at the single point 0. We generally do not want such isolated points of differentiability — this motivates the notion of holomorphicity, which demands differentiability throughout an open set.

CHAPTER 6

Holomorphic Functions

6.1 Definition

Definition 6.1: Holomorphic function

Let $U \subseteq \mathbb{C}$ be open, let $f : U \rightarrow \mathbb{C}$ be given, and $z_0 \in U$. We say f is **holomorphic at** z_0 if there exists a neighbourhood of z_0 where f is differentiable. We say f is **holomorphic on** U if f is holomorphic at every point of U — equivalently, if f is differentiable on U .

Throughout, D denotes a **domain**: an open and connected subset of $\mathbb{C}$. We write

$$\mathcal{H}(D) = \{\text{holomorphic functions on } D\}.$$

6.2 Property 1: zero derivative forces constancy

Proposition 6.1

Let D be a domain and $f \in \mathcal{H}(D)$ with $f' = 0$ on D . Then f is constant.

The key is the following real-variable lemma.

Lemma 6.1

Let D be a domain and $g : D \rightarrow \mathbb{R}$ be such that g_x, g_y vanish identically on D . Then g is constant on D .

Proof of Proposition (using the Lemma). Let $z_0 \in D$. Since $f'(z_0) = f_x(z_0) = u_x(z_0) + iv_x(z_0) = 0$, we get $u_x(z_0) = 0$ and $v_x(z_0) = 0$. By the Cauchy–Riemann equations, $v_y = u_x = 0$ and $u_y = -v_x = 0$. Hence

$$u_x \equiv 0, u_y \equiv 0 \quad \text{and} \quad v_x \equiv 0, v_y \equiv 0 \quad \text{on } D.$$

By the Lemma, u is constant and v is constant, so $f = u + iv$ is constant. ■

Proof of the Lemma. Fix two points and join them by horizontal/vertical segments inside D (possible since D is open and connected, hence stair-connected). On a horizontal segment from (a, b) to (a', b) , the Mean Value Theorem gives, for some a'' between a and a' ,

$$\frac{g(a', b) - g(a, b)}{a' - a} = g_x(a'', b) = 0 \implies g(a', b) = g(a, b).$$

Similarly along a vertical segment from (a', b) to (a', b') , for some b'' between b and b' ,

$$\frac{g(a', b') - g(a', b)}{b' - b} = g_y(a', b'') = 0 \implies g(a', b') = g(a', b).$$

Chaining finitely many such segments joins any two points of D with equal g -values, so g is constant. ■

6.3 How to check holomorphicity

Given $f = u + iv$, the practical test combines the Cauchy–Riemann equations with continuity:

- If the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$ fail at a point, f is *not* holomorphic there.
- If u_x, u_y, v_x, v_y are continuous *and* the Cauchy–Riemann equations hold, then f is holomorphic (by the partial-converse theorem).

6.3.1 Harmonic functions

If u is the real part of a holomorphic function then, differentiating the Cauchy–Riemann equations,

$$u_{xx} + u_{yy} = 0,$$

i.e. u is **harmonic**. (From $u_x = v_y$ and $u_y = -v_x$: $u_{xx} = v_{yx}$ and $u_{yy} = -v_{xy}$, which cancel by equality of mixed partials.)

6.4 Examples of constancy

Example 6.1: Constant real part $\Rightarrow$ constant function

Let D be a domain and $f = u + iv \in \mathcal{H}(D)$. Prove: if u (the real part) is constant, then f is constant.

Solution. u constant gives $u_x = 0$, $u_y = 0$. By Cauchy–Riemann, $v_y = u_x = 0$ and $v_x = -u_y = 0$. Hence u and v are both constant, so $f = u + iv$ is constant.

Corollary 6.1

A holomorphic function $f : \mathbb{C} \rightarrow \mathbb{R}$ must be constant. (Its imaginary part is $\equiv 0$, hence constant, so by the previous example f is constant.)

Example 6.2: Constant modulus $\Rightarrow$ constant function

Let D be a domain and $f = u + iv \in \mathcal{H}(D)$ with $|f|$ constant. Then f is constant.

Solution. Write $|f|^2 = u^2 + v^2$. *Case* $|f|^2 = 0$: then $u^2 + v^2 = 0$, so $u = 0$ and $v = 0$, and f is constant. *Case* $|f|^2 = c$ (*const*): differentiating $u^2 + v^2 = c$,

$$uu_x + vv_x = 0, \quad uu_y + vv_y = 0.$$

Using Cauchy–Riemann $u_x = v_y$, $u_y = -v_x$:

$$uu_x - vu_y = 0 \quad (1), \quad uu_y + vu_x = 0 \quad (2).$$

From (1), $u_x = \frac{v}{u}u_y$ (assuming $u \neq 0$); substitute into (2):

$$uu_y + \frac{v^2}{u}u_y = 0 \implies u_y \left(\frac{u^2 + v^2}{u} \right) = 0 \implies u_y = 0,$$

since $\frac{u^2 + v^2}{u} \neq 0$. Similarly $u_x = 0$. Hence u is constant, and therefore f is constant. (The case $u \equiv 0$ is handled symmetrically using v .)

CHAPTER 7

Power Series

7.1 Formal power series

Definition 7.1: Formal power series

A **formal power series** is an expression of the form

$$\sum_{n \geq 0} a_n x^n, \quad a_n \in F, \quad x \text{ an indeterminate.}$$

The motivating case is the geometric series $a_n = 1$, i.e. $\sum_{n \geq 0} x^n$. For $z \in \mathbb{C}$, the partial sums are

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z}.$$

If $|z| < 1$, then $|z|^{n+1} \rightarrow 0$, so the series converges:

$$\sum_{n \geq 0} z^n = \frac{1}{1 - z}, \quad |z| < 1.$$

If $|z| > 1$, then $|z|^{n+1} \rightarrow \infty$ and the series does not converge. There is a *threshold* radius beyond which the power series diverges and below which it converges.

7.2 Radius of convergence

Definition 7.2: Radius of convergence, disc of convergence

For $\sum_{n \geq 0} a_n x^n$ set

$$S = \left\{ r \geq 0 : \sum_{n \geq 0} |a_n| r^n < \infty \right\}.$$

Since $0 \in S$, S is nonempty, and S is an interval of the form $[0, \infty)$, $[0, \rho]$ or $[0, \rho)$. Define ρ to be $\sup S$ if it exists, and otherwise set $\rho = \infty$. The quantity ρ is the **radius of convergence** of the formal power series, and

$$\{z \in \mathbb{C} : |z| < \rho\}$$

is called the **disc of convergence**.

Example 7.1: $\sum r^n$

$S = \{r \geq 0 : \sum r^n < \infty\} = [0, 1)$, so $\sup S = 1$ and $\rho = 1$.

7.3 Normal convergence

Definition 7.3: Normal convergence

Let $f_n : X \rightarrow \mathbb{C}$ ($n \geq 1$) be a sequence of functions on a set X . Define

$$\|f_n\|_X = \sup_{x \in X} |f_n(x)|.$$

We say $(f_n)_{n \geq 1}$ is **normally summable** (or $\sum f_n$ **converges normally**) on X if

$$\sum_{n \geq 1} \|f_n\|_X < \infty.$$

Example 7.2: Consequences of normal convergence

(1) Normal convergence of $\sum_{n \geq 1} f_n$ implies *absolute* convergence of $\sum_{n \geq 1} f_n(x)$ for every $x \in X$: indeed $\sum \sup |f_n|$ converges and $|f_n(x)| \leq \sup |f_n|$, hence $\sum |f_n(x)|$ converges.

(2) Normal convergence implies *uniform* convergence of $\sum_{n \geq 1} f_n$ (this is the **Weierstrass M-test**).

(3) The converse of (2) fails: uniform convergence does *not* imply normal convergence. Take

$$f_n(x) = \frac{x^n}{n}, \quad 0 \leq x < 1.$$

On $[0, a]$ with $0 < a < 1$ we have $|\frac{x^n}{n}| \leq \frac{a^n}{n}$, and since $\sum \frac{a^n}{n}$ converges (comparison with the geometric series), the Weierstrass M-test gives that $\sum \frac{x^n}{n}$ converges uniformly on $[0, a]$. But on $X = [0, 1)$,

$$\sup_{[0,1)} \frac{x^n}{n} = \frac{1}{n}, \quad \sum \frac{1}{n} = \infty,$$

so the series is *not* normally convergent on $[0, 1)$.

7.4 Convergence of a power series inside its disc

Theorem 7.1: Normal convergence on compact subdiscs

Consider a formal power series $\sum_{n \geq 0} a_n x^n$ with radius of convergence ρ . Then for any $r < \rho$ the series $\sum_{n \geq 0} a_n z^n$ converges *normally* on $\{z : |z| \leq r\}$. The series does not converge for $|z| > \rho$.

The engine of the proof is Abel's lemma.

Lemma 7.1: Abel's Lemma

Let $0 < r < r_0$. Suppose there exists $M > 0$ such that $|a_n| r_0^n \leq M$ for all sufficiently large n . Then $\sum_{n \geq 0} a_n x^n$ converges normally on $D = \{|z| \leq r\}$.

Proof of Abel's Lemma. On D ,

$$\|a_n z^n\|_D \leq |a_n| r^n = |a_n| r_0^n \left(\frac{r}{r_0}\right)^n \leq M \left(\frac{r}{r_0}\right)^n \quad \text{for large enough } n.$$

Since $\frac{r}{r_0} < 1$, summing the geometric bound,

$$\sum_{n \geq 0} \|a_n z^n\|_D \leq \sum_{n \geq 0} M \left(\frac{r}{r_0} \right)^n = M \cdot \frac{1}{1 - \frac{r}{r_0}} < \infty.$$

Hence $\sum_{n \geq 0} \|a_n z^n\|_D < \infty$, i.e. the series converges normally on D . ■

Proof of the Theorem. Let $r < \rho$ be given; pick r_0 with $0 \leq r < r_0 < \rho$. By definition of ρ , $\sum_{n \geq 0} |a_n| r_0^n < \infty$; in particular the general term is bounded, say $|a_n| r_0^n \leq M$ for all $n \geq 0$. Abel's Lemma now gives normal convergence on $\{|z| \leq r\}$.

For divergence when $|z| > \rho$: we show $(a_n z^n)$ is unbounded, hence $\sum a_n z^n$ cannot converge. Suppose, for contradiction, $|a_n z^n| = |a_n| |z|^n \leq M$ for all n . Choose R with $\rho < R < |z|$. Then

$$|a_n| R^n = |a_n| |z|^n \left(\frac{R}{|z|} \right)^n \leq M \left(\frac{R}{|z|} \right)^n,$$

so $\sum_{n \geq 0} |a_n| R^n < \infty$, forcing $R \in S$ with $\rho < R$ — contradicting $\rho = \sup S$. ■

7.5 Formulas for the radius of convergence

Theorem 7.2: Hadamard's formula

For a formal power series $\sum_{n \geq 0} a_n x^n$, the radius of convergence ρ is given by

$$\frac{1}{\rho} = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

Theorem 7.3: Ratio formula

For $\sum_{n \geq 0} a_n x^n$, suppose $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$ exists in $[0, \infty) \cup \{\infty\}$. Then

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|.$$

7.6 Differentiability of a power series

Inside the disc of convergence a power series defines an infinitely differentiable function, and one may differentiate term by term.

Theorem 7.4: Term-by-term differentiation

Let $\sum_{n \geq 0} a_n x^n$ be a formal power series with radius of convergence $\rho > 0$. Then:

- (1) The series $\sum_{n \geq k} a_n n(n-1) \cdots (n-k+1) x^{n-k}$ has radius of convergence ρ .
- (2) For any $a \in \mathbb{C}$, the function $f(z) = \sum_{n \geq 0} a_n (z-a)^n$ is infinitely differentiable in $B(a, \rho) =$

$\{z : |z - a| < \rho\}$, and

$$f^{(k)}(z) = \sum_{n \geq k} a_n n(n-1) \cdots (n-k+1) (z-a)^{n-k}$$

for all $k \geq 1$ and $|z - a| < \rho$.

(3) For every $n \geq 0$,

$$a_n = \frac{1}{n!} f^{(n)}(a).$$

Proof. (1) It suffices to treat $k = 1$: the series $\sum_{n \geq 0} a_n n x^{n-1}$ has radius ρ_0 with

$$\frac{1}{\rho_0} = \limsup_{n \rightarrow \infty} |n a_n|^{1/(n-1)}.$$

Since $\lim_{n \rightarrow \infty} n^{1/(n-1)} = 1$, one gets $\rho_0 = (\limsup_{n \rightarrow \infty} |a_n|^{1/(n-1)})^{-1} = \rho$. Concretely, if $0 \leq r < \rho_0$ then from $\sum |a_n| r^n < \infty$ one deduces $\sum |a_{n+1}| r^{n+1} < \infty$, hence $\sum |a_n| r^n < \infty$, giving $r \leq \rho$ and so $\rho_0 \leq \rho$; conversely if $0 \leq r < \rho$ then $r \sum |a_n| r^{n-1} \leq \sum |a_n| r^n < \infty$ shows $\sum |a_n| r^{n-1} < \infty$ and so $\sum |a_{n+1}| r^n < \infty$, giving $r \leq \rho_0$ and $\rho \leq \rho_0$. Thus $\rho_0 = \rho$.

(2) We prove the case $k = 1$; the general case follows by induction. Fix $w \in B(a, \rho)$; we show $f'(w) = g(w)$ where $g(z) = \sum_{n \geq 1} n a_n (z-a)^{n-1}$. Choose r with $|a - w| < r < \rho$. Write the partial sum $S_n(z) = \sum_{k=0}^n a_k (z-a)^k$ and tail $\sigma_n(z) = \sum_{k > n} a_k (z-a)^k$. For $z \neq w$,

$$\frac{f(z) - f(w)}{z - w} - g(w) = \underbrace{\left(\frac{S_n(z) - S_n(w)}{z - w} - S'_n(w) \right)}_{\text{partial-sum term}} + \underbrace{(S'_n(w) - g(w))}_{\text{2nd term}} + \underbrace{\frac{\sigma_n(z) - \sigma_n(w)}{z - w}}_{\text{3rd (remainder) term}}.$$

Third term. Using $(z-a)^k - (w-a)^k = ((z-a) - (w-a)) \sum \cdots$,

$$\frac{\sigma_n(z) - \sigma_n(w)}{z - w} = \frac{1}{z - w} \sum_{k > n} a_k \{ (z-a)^k - (w-a)^k \},$$

and each bracket satisfies

$$\left| \frac{(z-a)^k - (w-a)^k}{(z-a) - (w-a)} \right| = |(z-a)^{k-1} + (z-a)^{k-2}(w-a) + \cdots + (w-a)^{k-1}| \leq k r^{k-1}.$$

Hence

$$|\text{3rd term}| \leq \sum_{k > n} |a_k| k r^{k-1},$$

the tail of the convergent series $\sum_{k \geq 1} k a_k x^{k-1}$ (radius $\rho > r$ by (1)). Given $\varepsilon > 0$, choose n_0 with $|\text{3rd term}| < \varepsilon/3$ for $n \geq n_0$.

Second term. $S'_n(w) = \sum_{k=1}^n k a_k (w-a)^{k-1} \rightarrow g(w)$ as $n \rightarrow \infty$, so choose n_1 with $|S'_n(w) - g(w)| < \varepsilon/3$ for $n \geq n_1$. Put $n_\varepsilon = \max(n_0, n_1)$; then $|\text{2nd}| + |\text{3rd}| < 2\varepsilon/3$.

Partial-sum (first) term. S_{n_ε} is a polynomial, hence differentiable at w ; choose $\delta > 0$ with $|\frac{S_{n_\varepsilon}(z) - S_{n_\varepsilon}(w)}{z - w} - S'_{n_\varepsilon}(w)| < \varepsilon/3$ for $z \in B(w, \delta)$. Then for all such z ,

$$\left| \frac{f(z) - f(w)}{z - w} - g(w) \right| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $f'(w) = g(w)$.

(3) From (2), $f^{(k)}(z) = \sum_{n \geq k} a_n n(n-1) \cdots (n-k+1)(z-a)^{n-k}$. Put $z = a$: all terms with $n > k$ vanish, leaving

$$f^{(k)}(a) = a_k k(k-1) \cdots 1 = k! a_k, \quad \text{i.e.} \quad a_k = \frac{f^{(k)}(a)}{k!}. \quad \blacksquare$$

Corollary 7.1

Let $\sum_{n \geq 0} a_n x^n$ be a formal power series with radius of convergence $\rho > 0$. Then for any $a \in \mathbb{C}$, the function $f(z) = \sum_{n \geq 0} a_n (z-a)^n$ defines a holomorphic function on $B(a, \rho)$.

Example 7.3: The exponential series

Consider $\sum_{n \geq 0} \frac{x^n}{n!}$, so $a_n = \frac{1}{n!}$. By the ratio formula,

$$\left| \frac{a_n}{a_{n+1}} \right| = \frac{(n+1)!}{n!} = n+1 \rightarrow \infty,$$

so $\rho = +\infty$: the series converges for every $z \in \mathbb{C}$. For $z \in \mathbb{C}$ the function

$$e^z := \sum_{n \geq 0} \frac{z^n}{n!}$$

is holomorphic on all of $\mathbb{C}$ (the **exponential function**).

CHAPTER 8

The Elementary Functions

Definition 8.1: Entire function

A function which is holomorphic on all of $\mathbb{C}$ is called an **entire function**.

Examples: polynomials and the exponential function.

8.1 The exponential function

We have $e^z = \sum_{n \geq 0} \frac{z^n}{n!}$, and its derivative is again itself:

$$\frac{d}{dz} e^z = \sum_{n \geq 1} n \frac{z^{n-1}}{n!} = \sum_{n \geq 0} \frac{z^n}{n!} = e^z.$$

Theorem 8.1: Addition law

$e^{z_1+z_2} = e^{z_1} e^{z_2}$ for all $z_1, z_2 \in \mathbb{C}$.

Proof. Fix z_2 and set $g(z) = e^z e^{z_2-z}$. Then

$$g'(z) = e^z e^{z_2-z} - e^z e^{z_2-z} = 0,$$

so g is constant. Hence $g(z) = g(0) = e^0 e^{z_2} = e^{z_2}$; that is $e^z e^{z_2-z} = e^{z_2}$. Putting $z = z_1$ gives $e^{z_1} e^{z_2-z_1} = e^{z_2}$, and replacing z_2 by $z_1 + z_2$ yields $e^{z_1} e^{z_2} = e^{z_1+z_2}$. ■

Remark

The coefficients $\frac{1}{n!}$ are real, so $\overline{e^z} = e^{\bar{z}}$. Hence

$$|e^z|^2 = e^z \overline{e^z} = e^z e^{\bar{z}} = e^{z+\bar{z}} = e^{2\operatorname{Re}(z)}, \quad \text{so} \quad |e^z| = e^{\operatorname{Re}(z)}.$$

Writing $z = x + iy$, $e^z = e^x e^{iy} = |e^z| e^{iy}$; thus e^z takes all complex values *except* 0 (its modulus e^x is never 0).

! Caution

The real exponential is injective; the complex exponential is NOT. Since $e^{iy} = e^{i(y+2\pi)}$, we have

$$e^z = e^{z+2\pi i n}, \quad n \in \mathbb{Z}.$$

The value repeats on every horizontal strip of height 2π .

8.2 Cosine and sine

Define, as power series,

$$\begin{aligned}\cos z &= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots + (-1)^n \frac{z^{2n}}{(2n)!} + \cdots, \\ \sin z &= z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots + (-1)^n \frac{z^{2n+1}}{(2n+1)!} + \cdots.\end{aligned}$$

(Both are entire; the radius of convergence is ∞ , using $\lim_{n \rightarrow \infty} (n!)^{1/n} = \infty$.) Term-by-term differentiation gives

$$(\cos z)' = -\sin z, \quad (\sin z)' = \cos z.$$

In terms of the exponential,

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i},$$

and the fundamental identity holds for all complex z :

$$\cos^2 z + \sin^2 z = 1 \quad \forall z \in \mathbb{C}.$$

8.3 Logarithmic functions

We try to find a function $f(z)$ such that

$$e^{f(z)} = z.$$

Remark

The exponential function never takes the value 0, so f can be defined only for *nonzero* z . Moreover the choice of f is *not unique*: if $e^{f(z)} = z$ then $e^{f(z)+2\pi in} = z$ as well, for every $n \in \mathbb{Z}$.

Suppose $f(z) = u(z) + iv(z)$. From $e^{f(z)} = e^{u(z)}e^{iv(z)} = z = |z|e^{i\theta}$ we read off

$$e^{u(z)} = |z| \implies u(z) = \log |z| \text{ (real logarithm),}$$

$$v(z) = \theta + 2\pi n = \text{Arg}(z) + 2\pi n, \quad n \in \mathbb{Z}.$$

Hence

$$f(z) = \log |z| + i(\text{Arg}(z) + 2\pi n), \quad n \in \mathbb{Z},$$

where the principal argument $\text{Arg}(z) \in (-\pi, \pi]$.

Taking $n = 0$ gives $f(z) = \log |z| + i \text{Arg}(z)$. We check whether f is continuous.

! Caution

$\text{Arg}(z)$ is *not continuous* on the negative real axis: approaching from above the argument tends to π , from below to $-\pi$. Restricting the argument to a range $(\theta, \theta + 2\pi]$ removes the jump but always leaves one ray of discontinuity.

Definition 8.2: Branch of the logarithm

We say that a continuous function $f(z)$, defined on a domain D with $0 \notin D$, is a **branch of log on D** if for all $z \in D$,

$$e^{f(z)} = z.$$

Example 8.1: A branch on a slit plane

On $D = \{re^{i\theta} : r > 0, \theta \in (\alpha, \alpha + 2\pi)\}$ (with $\alpha \in \mathbb{R}$),

$$f(re^{i\theta}) = \log r + i\theta.$$

The **principal branch** of the logarithm is the case $D = \{re^{i\theta} : r > 0, \theta \in (-\pi, \pi)\}$, with

$$\log(re^{i\theta}) = \log r + i\theta.$$

Proposition 8.1: Branches differ by $2\pi in$

Let f be a branch of log on a domain D . Then any other branch g of log on D looks like

$$g(z) = f(z) + 2\pi in, \quad n \in \mathbb{Z} \text{ (fixed).}$$

Proof. Let g be another branch on D . For $z \in D$, $e^{f(z)} = z = e^{g(z)}$, so $e^{f(z)-g(z)} = 1$, whence $f(z) - g(z) \in 2\pi i\mathbb{Z}$ for each z . Thus $f(z) - g(z)$ takes values in the set $2\pi i\mathbb{Z}$. But $f - g$ is continuous on the connected set D , so its image is connected, and the only connected subsets of the discrete set $2\pi i\mathbb{Z}$ are singletons. Therefore $f(z) - g(z)$ is constant, say $= 2\pi in_0$, and $g(z) = f(z) - 2\pi in_0$. ■

8.4 Continuity of a branch gives holomorphicity

Theorem 8.2: Holomorphic inverse

Let U, V be open subsets of $\mathbb{C}$, let $f : U \rightarrow \mathbb{C}$ and $g : V \rightarrow \mathbb{C}$ be functions with $f(U) \subseteq V$ and $g(f(z)) = z$ for all $z \in U$. If g is holomorphic on V and g' does not vanish, then f is holomorphic on U and

$$f'(z) = \frac{1}{g'(f(z))}.$$

Corollary 8.1: Branches of log are holomorphic

A branch of log is holomorphic, with

$$f'(z) = \frac{1}{e^{f(z)}} = \frac{1}{z}.$$

Here g is the exponential function, which is holomorphic and non-vanishing.

Proof of the Theorem. Since $g \circ f = \text{id}$ on U , f is injective. Let $a \in U$; we show f is differentiable

at a , i.e. $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$ exists and equals $\frac{1}{g'(f(a))}$. Write

$$\frac{f(a+h)-f(a)}{h} = \frac{f(a+h)-f(a)}{g(f(a+h))-g(f(a))} = \frac{1}{\frac{g(f(a+h))-g(f(a))}{f(a+h)-f(a)}},$$

using $g(f(a+h))-g(f(a)) = (a+h)-a = h$. As $h \rightarrow 0$, continuity of f gives $\alpha_h := f(a+h)-f(a) \rightarrow 0$, and

$$\frac{g(f(a)+\alpha_h)-g(f(a))}{\alpha_h} \rightarrow g'(f(a)) \neq 0.$$

Therefore

$$\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = \frac{1}{g'(f(a))}. \quad \blacksquare$$

Remark

Summary for the logarithm. $e^{f(z)} = z$ needs a domain avoiding 0; $f(z) = \log |z| + i \arg(z)$ with $\arg(z) \in (\theta, \theta+2\pi)$ or, for the principal branch, $\text{Arg}(z) \in (-\pi, \pi]$; continuity of a branch $\Rightarrow$ holomorphicity, with $f'(z) = 1/z$.

8.5 The power series of $\log(1+z)$

Since $\frac{d}{dz} \log(1+w) = \frac{1}{1+w}$, and for $|w| < 1$

$$\frac{1}{1+w} = 1 - w + w^2 - w^3 + w^4 - \dots,$$

we look for a formal power series whose derivative is the above; it is

$$w - \frac{w^2}{2} + \frac{w^3}{3} - \frac{w^4}{4} + \dots, \quad \text{radius} = 1.$$

For the principal branch of $\log$,

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \dots, \quad |z| < 1.$$

Remark

This is a first instance of a general phenomenon proved later: **any holomorphic function has a power series expansion around each point of its domain**. If f is holomorphic around z_0 , then $f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$ near z_0 .

CHAPTER 9

Complex Integration

Complex integration is built on Riemann integration over a real interval $[a, b]$, transported to the plane along a path.

9.1 Paths

Definition 9.1: Path; smooth; piecewise smooth

Let U be an open subset of $\mathbb{C}$. By a **path in U** we mean a continuous function $\gamma : [a, b] \rightarrow U$ for some $a < b$ in $\mathbb{R}$. We call a path **smooth** if for each $t \in (a, b)$ the derivative $\gamma'(t)$ exists and $\gamma' : (a, b) \rightarrow \mathbb{C}$ is continuous. We call it **piecewise smooth** if there is a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ such that γ is smooth on each $[t_i, t_{i+1}]$.

Theorem 9.1: Length of a path

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise smooth path (so γ is of bounded variation). Then

$$\int_a^b |\gamma'(t)| dt \quad \text{is finite.}$$

This quantity is called the **length** of γ , written L_γ .

9.2 The Riemann–Stieltjes integral and the path integral

Theorem 9.2: Riemann–Stieltjes integral

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise smooth path and $f : [\gamma] \rightarrow \mathbb{C}$ continuous (on the image of γ). Then there is a complex number I such that for any $\varepsilon > 0$ there is $\delta > 0$ so that for every partition $a = t_0 < t_1 < \cdots < t_n = b$ with $|t_{k+1} - t_k| < \delta$ (and any tags $\tilde{t}_k$),

$$\left| I - \sum_{k=0}^{n-1} f(\tilde{t}_k) (\gamma(t_{k+1}) - \gamma(t_k)) \right| < \varepsilon.$$

The number I is called the integral of f with respect to γ over $[a, b]$; in fact

$$I = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Definition 9.2: Path integral

Let $\gamma : [a, b] \rightarrow U$ be a piecewise smooth path and $f : U \rightarrow \mathbb{C}$ continuous. The **(path) integral of f along γ** is

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

9.2.1 Linearity

$$\int_{\gamma} (cf(z) + g(z)) dz = c \int_{\gamma} f(z) dz + \int_{\gamma} g(z) dz.$$

9.3 The fundamental example**Example 9.1: $\int_{\gamma} z^m dz$ around the unit circle**

Let $\gamma : [0, 1] \rightarrow \mathbb{C}$, $\gamma(t) = e^{2\pi it}$ (the unit circle, positively oriented), and $f_m(z) = z^m$, $m \in \mathbb{Z}$. Then $\gamma'(t) = 2\pi i e^{2\pi it}$.

Case $m = -1$.

$$\int_{\gamma} \frac{1}{z} dz = \int_0^1 \frac{1}{e^{2\pi it}} (2\pi i e^{2\pi it}) dt = \int_0^1 2\pi i dt = 2\pi i.$$

General $m \neq -1$.

$$\begin{aligned} \int_{\gamma} z^m dz &= \int_0^1 e^{2\pi imt} (2\pi i e^{2\pi it}) dt = 2\pi i \int_0^1 e^{2\pi i(m+1)t} dt \\ &= 2\pi i \int_0^1 [\cos 2\pi(m+1)t + i \sin 2\pi(m+1)t] dt = 0, \end{aligned}$$

because both $\int_0^1 \cos 2\pi(m+1)t dt$ and $\int_0^1 \sin 2\pi(m+1)t dt$ vanish for $m \neq -1$. Hence

$$\int_{\gamma} z^m dz = \begin{cases} 0, & m \neq -1, \\ 2\pi i, & m = -1, \end{cases} \quad \gamma(t) = e^{2\pi it}, \quad t \in [0, 1].$$

Remark

This single computation is the seed of the entire residue theory: only the z^{-1} term contributes a nonzero integral around a loop enclosing 0.

9.4 Basic properties of the path integral

Definition 9.3: Equivalent paths

Two smooth paths $\gamma_1 : [a, b] \rightarrow \mathbb{C}$ and $\gamma_2 : [c, d] \rightarrow \mathbb{C}$ are called **equivalent** if there is a one-one C^1 function $\lambda : [c, d] \rightarrow [a, b]$ with $\lambda(c) = a$, $\lambda(d) = b$ and $\gamma_2 = \gamma_1 \circ \lambda$. (Intuitively, they trace the same curve with a different parameter — a different speed, but the same direction.) We write $\gamma_1 \sim \gamma_2$.

Theorem 9.3: Invariance under reparametrisation

If $\gamma_1 \sim \gamma_2$, then $\int_{\gamma_1} f dz = \int_{\gamma_2} f dz$.

Proof. Write $f = u + iv$ and $\gamma_j = \alpha_j + i\beta_j$. By definition,

$$\int_{\gamma_2} f dz = \int_c^d f(\gamma_2(t)) \gamma_2'(t) dt.$$

Using $\gamma_2 = \gamma_1 \circ \lambda$ and $\gamma_2'(t) = \gamma_1'(\lambda(t))\lambda'(t)$, the substitution $s = \lambda(t)$ (so $ds = \lambda'(t) dt$, with $\lambda(c) = a$, $\lambda(d) = b$) gives

$$\int_c^d f(\gamma_2(t)) \gamma_2'(t) dt = \int_a^b f(\gamma_1(s)) \gamma_1'(s) ds = \int_{\gamma_1} f dz.$$

(The change of variables is applied to each of the real integrals coming from the expansion into $u, v, \alpha_j', \beta_j'$.) ■

Definition 9.4: Reverse path

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a path. Define $\tilde{\gamma} : [a, b] \rightarrow \mathbb{C}$ by $\tilde{\gamma}(t) = \gamma(b + a - t)$. Then $\tilde{\gamma}$ is the **reverse path** (it traverses the same curve from $\gamma(b)$ back to $\gamma(a)$).

Proposition 9.1

$$\int_{\tilde{\gamma}} f(z) dz = - \int_{\gamma} f(z) dz.$$

Proof. $\int_{\tilde{\gamma}} f dz = \int_a^b f(\tilde{\gamma}(t)) \tilde{\gamma}'(t) dt = - \int_a^b f(\gamma(b + a - t)) \gamma'(b + a - t) dt$. Substituting $\lambda(t) = b + a - t$ (so $\lambda'(t) = -1$, $\lambda(a) = b$, $\lambda(b) = a$) turns this into $-\int_b^a f(\gamma(s)) \gamma'(s) ds = -\int_{\gamma} f dz$, using $\int_b^a = -\int_a^b$. ■

9.5 The ML inequality

Recall for a real interval $|\int_a^b f(t) dt| \leq \int_a^b |f(t)| dt$. The analogue for path integrals is the following (note one *cannot* naively write $\int_{\gamma} |f(z)| dz$, which is meaningless as a path integral; the correct bound uses $\int_a^b |f(\gamma(t))| |\gamma'(t)| dt$).

Theorem 9.4: ML inequality

Let $\gamma : [a, b] \rightarrow U$ be piecewise smooth and $f : U \rightarrow \mathbb{C}$ continuous. Then

$$\left| \int_{\gamma} f(z) dz \right| \leq M_{\gamma} L_{\gamma},$$

where $M_{\gamma} = \sup_{t \in [a, b]} |f(\gamma(t))|$ and $L_{\gamma} = \int_a^b |\gamma'(t)| dt$ is the length of γ .

Proof. We first show the key inequality ($\star$):

$$\left| \int_{\gamma} f(z) dz \right| \leq \int_a^b |f(\gamma(t)) \gamma'(t)| dt. \quad (\star)$$

Granting ($\star$),

$$\left| \int_{\gamma} f dz \right| \leq \int_a^b |f(\gamma(t))| |\gamma'(t)| dt \leq M_{\gamma} \int_a^b |\gamma'(t)| dt = M_{\gamma} L_{\gamma}.$$

Proof of ($\star$). Let $g : [a, b] \rightarrow \mathbb{C}$ be continuous; we show $\left| \int_a^b g(t) dt \right| \leq \int_a^b |g(t)| dt$. Write $\int_a^b g(t) dt = R e^{i\theta}$ with $R \geq 0$. Then

$$R = \left| \int_a^b g(t) dt \right| = e^{-i\theta} \int_a^b g(t) dt = \int_a^b e^{-i\theta} g(t) dt.$$

Write $e^{-i\theta} g(t) = f_1(t) + i f_2(t)$ with f_1, f_2 real. Since R is real, its imaginary part is 0:

$$R = \int_a^b f_1(t) dt + i \int_a^b f_2(t) dt \implies \int_a^b f_2(t) dt = 0,$$

so

$$R = \int_a^b f_1(t) dt = \int_a^b \operatorname{Re}(e^{-i\theta} g(t)) dt.$$

Using $\operatorname{Re}(\zeta) \leq |\operatorname{Re}(\zeta)| \leq |\zeta|$,

$$R \leq \int_a^b |e^{-i\theta} g(t)| dt = \int_a^b |g(t)| dt.$$

Applying this with $g(t) = f(\gamma(t))\gamma'(t)$ gives ($\star$). ■

Corollary 9.1: Uniform limits and integration

Let f_n be a sequence of continuous functions $U \rightarrow \mathbb{C}$ converging *uniformly* to f . Then

$$\int_{\gamma} f(z) dz = \lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz, \quad \text{i.e.} \quad \int_{\gamma} \left(\lim_{n \rightarrow \infty} f_n(z) \right) dz = \lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz.$$

Proof. By the ML inequality,

$$\left| \int_{\gamma} f_n dz - \int_{\gamma} f dz \right| = \left| \int_{\gamma} (f_n - f) dz \right| \leq \sup_{[\gamma]} |f_n - f| \cdot L_{\gamma}.$$

Given $\varepsilon > 0$, uniform convergence gives n_0 with $\|f_n - f\|_{\infty} < \varepsilon$ for $n \geq n_0$, so the difference is $< \varepsilon L_{\gamma}$ for $n \geq n_0$. ■

9.6 The Fundamental Theorem of Calculus

Theorem 9.5: FTC in real analysis

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and $F(t) = \int_a^t f(x) dx$. Then F is (uniformly) continuous on $[a, b]$, differentiable on (a, b) with $F' = f$, and

$$\int_a^b f(x) dx = F(b) - F(a).$$

Theorem 9.6: FTC in complex analysis

Let $U \subseteq \mathbb{C}$ be open and $\gamma : [a, b] \rightarrow U$ piecewise smooth. If $f : U \rightarrow \mathbb{C}$ is continuous and has a **primitive** F (i.e. $F' = f$ on U), then

$$\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a)).$$

Proof. Using $F' = f$ and the chain rule $\frac{d}{dt}F(\gamma(t)) = F'(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t)$,

$$\int_{\gamma} f dz = \int_a^b f(\gamma(t))\gamma'(t) dt = \int_a^b (F \circ \gamma)'(t) dt = F(\gamma(b)) - F(\gamma(a)). \quad \blacksquare$$

! Caution

Not every continuous function on an open subset of $\mathbb{C}$ has a primitive. Consider $f(z) = x^2 + y^2 = |z|^2$, continuous on $\mathbb{C}$.

Claim. There is no F holomorphic on $\mathbb{C}$ with $F' = f$. Suppose $F = u + iv$ existed. By Cauchy–Riemann $u_x = v_y$, $u_y = -v_x$, and $F' = F_x = u_x + iv_x = f = x^2 + y^2$. As $F' = f$ is real-valued, $v_x = 0$, hence $u_y = -v_x = 0$, so u is a function of x alone. But then $u_x = x^2 + y^2$ would depend on y — a **contradiction**.

9.7 Closed curves

Definition 9.5: Closed and simple closed curves

A path $\gamma : [a, b] \rightarrow \mathbb{C}$ is a **closed curve** if $\gamma(a) = \gamma(b)$. It is a **simple closed curve** if in addition γ is injective on $[a, b)$.

For instance $\gamma(t) = e^{2\pi it}$, $t \in [0, 1]$, is simple closed, while $\gamma(t) = e^{4\pi it}$, $t \in [0, 1]$, is closed but *not* simple (it wraps twice).

Corollary 9.2: Primitive $\Rightarrow$ zero integral over closed curves

If, in the FTC, γ is closed, then $\int_{\gamma} f(z) dz = 0$.

(Indeed $F(\gamma(b)) - F(\gamma(a)) = 0$.) This is consistent with the fundamental example: $\int_{\gamma} z^m dz = 0$ for $m \neq -1$ because z^m has the primitive $\frac{z^{m+1}}{m+1}$, whereas z^{-1} has *no* primitive on $\mathbb{C} \setminus \{0\}$, which is exactly why its loop integral is $2\pi i \neq 0$.

9.8 Orientation

Definition 9.6: Positive orientation

A closed curve is **positively oriented** (traversed anticlockwise) if the region enclosed lies to the *left* as we trace the curve for increasing values of t .

CHAPTER 10

Cauchy's Theorem and the Integral Formula

The central discovery of the subject: the integral of a holomorphic function around a closed curve vanishes, and the values of a holomorphic function inside a curve are completely determined by its values on the curve.

10.1 The triangle (Goursat) theorem

Theorem 10.1: Goursat / Triangle theorem

Let $U \subseteq \mathbb{C}$ be open and $f \in \mathcal{H}(U)$. Let $\Delta \subseteq U$ be a (solid) triangle contained in U , and let $\partial\Delta$ be its boundary, positively oriented. Then

$$\int_{\partial\Delta} f(z) dz = 0.$$

Proof. Write $I = \int_{\partial\Delta} f dz$ and let L be the perimeter of Δ .

Subdivision. Join the midpoints of the three sides of Δ . This partitions Δ into four congruent sub-triangles $\Delta^{(1)}, \Delta^{(2)}, \Delta^{(3)}, \Delta^{(4)}$, each with perimeter $L/2$. The interior edges are each traversed twice in opposite directions, so their contributions cancel and

$$I = \int_{\partial\Delta} f dz = \sum_{j=1}^4 \int_{\partial\Delta^{(j)}} f dz.$$

Hence for at least one j , $|\int_{\partial\Delta^{(j)}} f dz| \geq \frac{|I|}{4}$. Call that sub-triangle Δ_1 ; then

$$\left| \int_{\partial\Delta_1} f dz \right| \geq \frac{|I|}{4}, \quad \text{perim}(\Delta_1) = \frac{L}{2}, \quad \text{diam}(\Delta_1) = \frac{\text{diam } \Delta}{2}.$$

Iteration and nesting. Repeat, producing nested triangles $\Delta \supseteq \Delta_1 \supseteq \Delta_2 \supseteq \cdots$ with

$$\left| \int_{\partial\Delta_n} f dz \right| \geq \frac{|I|}{4^n}, \quad \text{perim}(\Delta_n) = \frac{L}{2^n}, \quad \text{diam}(\Delta_n) = \frac{d}{2^n} \quad (d = \text{diam } \Delta).$$

The Δ_n are nonempty, compact and nested with $\text{diam } \Delta_n \rightarrow 0$; by **Cantor's intersection theorem** there is a unique point $z_0 \in \bigcap_{n \geq 1} \Delta_n \subseteq U$.

Using differentiability at z_0 . Since f is differentiable at z_0 , given $\varepsilon > 0$ there is $\delta > 0$ with

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon |z - z_0| \quad \text{for } |z - z_0| < \delta.$$

Choose n large enough that $\Delta_n \subseteq B(z_0, \delta)$. The functions $z \mapsto f(z_0)$ and $z \mapsto f'(z_0)(z - z_0)$ are polynomials, hence have primitives, so their integrals over the closed curve $\partial\Delta_n$ vanish:

$$\int_{\partial\Delta_n} [f(z_0) + f'(z_0)(z - z_0)] dz = 0.$$

Therefore

$$\int_{\partial\Delta_n} f dz = \int_{\partial\Delta_n} [f(z) - f(z_0) - f'(z_0)(z - z_0)] dz.$$

By the ML inequality, with $M_n = \sup_{z \in \partial\Delta_n} |z - z_0| \leq \text{diam } \Delta_n = d/2^n$ and length $L/2^n$,

$$\left| \int_{\partial\Delta_n} f dz \right| \leq \varepsilon \cdot \frac{d}{2^n} \cdot \frac{L}{2^n} = \frac{\varepsilon dL}{4^n}.$$

Combining with the lower bound $\frac{|I|}{4^n} \leq \left| \int_{\partial\Delta_n} f dz \right|$,

$$\frac{|I|}{4^n} \leq \frac{\varepsilon dL}{4^n} \implies |I| \leq \varepsilon dL.$$

As $\varepsilon > 0$ is arbitrary, $|I| = 0$, i.e. $I = 0$. ■

10.2 From the triangle theorem to primitives

Theorem 10.2: Integral / Primitive theorem

Let f be holomorphic on a *convex* open set U (or a disc). Then f has a primitive F on U ; consequently $\int_{\gamma} f dz = 0$ for every closed path γ in U .

Sketch. Fix a base point $z_* \in U$ and define $F(z) = \int_{[z_*, z]} f(\zeta) d\zeta$ along the straight segment (which lies in U by convexity). For $z, z+h \in U$, the triangle with vertices $z_*, z, z+h$ lies in U , so by Goursat the integral around it is 0; this gives

$$F(z+h) - F(z) = \int_{[z, z+h]} f(\zeta) d\zeta.$$

Dividing by h and using continuity of f , $\frac{F(z+h)-F(z)}{h} \rightarrow f(z)$, so $F' = f$. By the FTC (complex form), $\int_{\gamma} f dz = 0$ for every closed γ . ■

Theorem 10.3: Closed-curve theorem

Let f be holomorphic on a disc D (more generally a convex/star-shaped domain). Then for every closed path γ in D ,

$$\int_{\gamma} f(z) dz = 0.$$

10.3 The Cauchy Integral Formula

Theorem 10.4: Cauchy Integral Formula

Let f be holomorphic on an open set containing the closed disc $\overline{B(a, r)}$, and let γ be the boundary circle $a + \gamma_r$ positively oriented. Then for every α with $|\alpha - a| < r$,

$$f(\alpha) = \frac{1}{2\pi i} \int_{a+\gamma_r} \frac{f(z)}{z - \alpha} dz.$$

Proof. Define

$$g(z) = \begin{cases} \frac{f(z) - f(\alpha)}{z - \alpha}, & z \neq \alpha, \\ f'(\alpha), & z = \alpha. \end{cases}$$

Then g is holomorphic on the disc except possibly at α , and is continuous at α ; in fact g is holomorphic on the whole disc (the singularity is removable, as justified later, or one applies the closed-curve theorem to the extended g). Hence by the closed-curve theorem $\int_{a+\gamma_r} g(z) dz = 0$, i.e.

$$\int_{a+\gamma_r} \frac{f(z) - f(\alpha)}{z - \alpha} dz = 0 \implies \int_{a+\gamma_r} \frac{f(z)}{z - \alpha} dz = f(\alpha) \int_{a+\gamma_r} \frac{dz}{z - \alpha}.$$

It remains to evaluate $\int_{a+\gamma_r} \frac{dz}{z - \alpha}$. Parametrising the circle and using the fundamental example (only the $(z - \alpha)^{-1}$ power contributes),

$$\int_{a+\gamma_r} \frac{dz}{z - \alpha} = 2\pi i.$$

Therefore $\int_{a+\gamma_r} \frac{f(z)}{z - \alpha} dz = 2\pi i f(\alpha)$, which is the claim. ■

10.4 Power series expansion of holomorphic functions

Corollary 10.1: Power series expansion

Let f be holomorphic on $B(a, R)$. Then f has a power series expansion

$$f(z) = \sum_{n \geq 0} a_n (z - a)^n \quad \text{for } |z - a| < R,$$

where the coefficients are

$$a_n = \frac{1}{2\pi i} \int_{a+\gamma_r} \frac{f(z)}{(z - a)^{n+1}} dz = \frac{f^{(n)}(a)}{n!}, \quad 0 < r < R.$$

In particular f is infinitely differentiable and

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{a+\gamma_r} \frac{f(z)}{(z - a)^{n+1}} dz.$$

Sketch. In the Cauchy Integral Formula expand the kernel as a geometric series: for z on the circle of radius r and $|\alpha - a| < r$,

$$\frac{1}{z - \alpha} = \frac{1}{(z - a) - (\alpha - a)} = \frac{1}{z - a} \cdot \frac{1}{1 - \frac{\alpha - a}{z - a}} = \sum_{n \geq 0} \frac{(\alpha - a)^n}{(z - a)^{n+1}},$$

which converges uniformly in z on the circle (since $|\frac{\alpha - a}{z - a}| = \frac{|\alpha - a|}{r} < 1$). Interchanging sum and integral (Corollary on uniform limits and integration),

$$f(\alpha) = \frac{1}{2\pi i} \int \frac{f(z)}{z - \alpha} dz = \sum_{n \geq 0} \left(\frac{1}{2\pi i} \int \frac{f(z)}{(z - a)^{n+1}} dz \right) (\alpha - a)^n = \sum_{n \geq 0} a_n (\alpha - a)^n.$$

Comparison with term-by-term differentiation gives $a_n = f^{(n)}(a)/n!$. ■

10.5 Analytic = holomorphic

Definition 10.1: Analytic function

A function $f : U \rightarrow \mathbb{C}$ is **analytic** at $a \in U$ if there is $r > 0$ and a power series $\sum_{n \geq 0} a_n(z-a)^n$ with $f(z) = \sum_{n \geq 0} a_n(z-a)^n$ for all $z \in B(a, r)$. It is analytic on U if analytic at every point of U .

Theorem 10.5: Holomorphic $\iff$ analytic

For $f : U \rightarrow \mathbb{C}$ on an open set U ,

$$f \text{ is holomorphic on } U \iff f \text{ is analytic on } U.$$

Proof. ($\Leftarrow$) A convergent power series is holomorphic inside its disc of convergence (term-by-term differentiation). ($\Rightarrow$) is exactly the power series expansion corollary above. ■

! Caution

This equivalence is special to the complex setting. In real analysis a function can be C^∞ (infinitely differentiable) yet *not* analytic. The standard example is

$$g(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Here g is C^∞ on $\mathbb{R}$ with $g^{(n)}(0) = 0$ for all n , so its Taylor series at 0 is identically 0, which does not equal g in any neighbourhood of 0. Thus g is C^∞ but not analytic. *No such pathology occurs for holomorphic functions.*

10.6 Morera's theorem

Theorem 10.6: Morera's theorem

Let $U \subseteq \mathbb{C}$ be open and $f : U \rightarrow \mathbb{C}$ continuous. If

$$\int_{\partial \Delta} f(z) dz = 0$$

for every triangle $\Delta \subseteq U$, then f is holomorphic on U .

Proof. Holomorphicity is local, so work in a disc $B \subseteq U$. As in the primitive theorem, fix a base point and define $F(z) = \int_{[z_*, z]} f(\zeta) d\zeta$. The hypothesis (vanishing of f over triangle boundaries) gives, for $z, z+h \in B$,

$$F(z+h) - F(z) = \int_{[z, z+h]} f(\zeta) d\zeta,$$

and by continuity of f , $\frac{F(z+h)-F(z)}{h} \rightarrow f(z)$, so $F' = f$ on B . Thus F is holomorphic on B ; being holomorphic, F is infinitely differentiable, hence $f = F'$ is holomorphic on B . As B was arbitrary, $f \in \mathcal{H}(U)$. ■

CHAPTER 11

Consequences of the Cauchy Theory

11.1 Uniform limits and the Cauchy estimates

Theorem 11.1: Uniform limit of holomorphic functions is holomorphic

Let $f_n \in \mathcal{H}(U)$ converge uniformly on compact subsets of U to f . Then $f \in \mathcal{H}(U)$, and moreover $f'_n \rightarrow f'$ uniformly on compact subsets.

Proof. f is continuous (uniform limit of continuous functions). For any triangle $\Delta \subseteq U$, by Goursat $\int_{\partial\Delta} f_n dz = 0$; letting $n \rightarrow \infty$ and using uniform convergence (ML inequality) gives $\int_{\partial\Delta} f dz = 0$. By Morera's theorem $f \in \mathcal{H}(U)$. ■

Theorem 11.2: Cauchy estimates

Let f be holomorphic on an open set containing $\overline{B(a, r)}$, and let $M = \sup_{|z-a|=r} |f(z)|$. Then for every $k \geq 0$,

$$|f^{(k)}(a)| \leq \frac{k! M}{r^k}.$$

Proof. From the integral formula for derivatives,

$$f^{(k)}(a) = \frac{k!}{2\pi i} \int_{a+\gamma_r} \frac{f(z)}{(z-a)^{k+1}} dz.$$

On the circle $|z-a|=r$ the integrand has modulus $\leq M/r^{k+1}$, and the length is $2\pi r$; by the ML inequality,

$$|f^{(k)}(a)| \leq \frac{k!}{2\pi} \cdot \frac{M}{r^{k+1}} \cdot 2\pi r = \frac{k! M}{r^k}. \quad \blacksquare$$

11.2 Liouville's theorem

Theorem 11.3: Liouville

A bounded entire function is constant.

Proof. Let f be entire with $|f| \leq M$ on $\mathbb{C}$. For any $a \in \mathbb{C}$ and any $r > 0$, the Cauchy estimate with $k = 1$ gives

$$|f'(a)| \leq \frac{1! M}{r} = \frac{M}{r}.$$

Letting $r \rightarrow \infty$, $|f'(a)| = 0$. So $f' \equiv 0$ on $\mathbb{C}$, and since $\mathbb{C}$ is a domain, f is constant. ■

Corollary 11.1: $\cos, \sin$ are unbounded on $\mathbb{C}$

$\cos$ and $\sin$ are entire and non-constant; if either were bounded on $\mathbb{C}$, Liouville would force it to be constant. Hence both are *unbounded* on $\mathbb{C}$ — in sharp contrast to the real case, where $|\cos x|, |\sin x| \leq 1$.

Theorem 11.4: Generalised Liouville

Let f be entire and suppose there exist constants $A, B \geq 0$ and an integer $n \geq 0$ with

$$|f(z)| \leq A + B |z|^n \quad \text{for all } z \in \mathbb{C}.$$

Then f is a polynomial of degree $\leq n$.

Proof. Expand $f(z) = \sum_{k \geq 0} a_k z^k$ (valid on all of $\mathbb{C}$). For $k > n$ and any $r > 0$, the Cauchy estimate gives

$$|a_k| = \frac{|f^{(k)}(0)|}{k!} \leq \frac{M(r)}{r^k}, \quad M(r) = \sup_{|z|=r} |f| \leq A + Br^n.$$

Thus $|a_k| \leq \frac{A + Br^n}{r^k} \rightarrow 0$ as $r \rightarrow \infty$ (since $k > n$). Hence $a_k = 0$ for all $k > n$, so f is a polynomial of degree $\leq n$. ■

11.3 The Fundamental Theorem of Algebra

Theorem 11.5: Fundamental Theorem of Algebra

Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_1 z + a_0$ with complex coefficients ($a_n \neq 0$, $n \geq 1$) has a root in $\mathbb{C}$.

Proof. Suppose, for contradiction, $P(z) \neq 0$ for all $z \in \mathbb{C}$. Then $g(z) = \frac{1}{P(z)}$ is entire. We show g is bounded. As $|z| \rightarrow \infty$,

$$|P(z)| = |z|^n \left| a_n + \frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n} \right| \rightarrow \infty,$$

so $|g(z)| = 1/|P(z)| \rightarrow 0$; in particular there is $R > 0$ with $|g(z)| \leq 1$ for $|z| \geq R$. On the compact disc $|z| \leq R$, g is continuous, hence bounded. So g is bounded on all of $\mathbb{C}$. By Liouville g is constant, hence P is constant — contradicting $n \geq 1$. Therefore P has a root. ■

Theorem 11.6: Entire functions blowing up at infinity are polynomials

If f is entire and $|f(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, then f is a polynomial.

Idea. The hypothesis shows f has finitely many zeros, and controls the growth so that the generalised Liouville argument applies to f divided by a polynomial vanishing at those zeros; one concludes f is a polynomial. ■

11.4 Zeros of holomorphic functions

Lemma 11.1: Zeros are isolated (finiteness on compacts)

Let $f \in \mathcal{H}(D)$, $f \not\equiv 0$, on a domain D . Then the zeros of f have no accumulation point in D ; consequently f has only finitely many zeros in any compact subset of D .

Sketch. If $f(a) = 0$ and $f \not\equiv 0$ near a , expand $f(z) = \sum_{n \geq m} a_n(z-a)^n$ with $a_m \neq 0$ (the first nonzero coefficient). Then $f(z) = (z-a)^m h(z)$ with $h(a) = a_m \neq 0$ and h holomorphic, so $h \neq 0$ in a neighbourhood of a ; hence a is an isolated zero. A compact set covered by such neighbourhoods contains only finitely many zeros. ■

Definition 11.1: Order / multiplicity of a zero

If $f(z) = (z-a)^m h(z)$ near a with $h(a) \neq 0$ and $m \geq 1$, we say a is a zero of f of **order** (or **multiplicity**) m . Equivalently $f(a) = f'(a) = \dots = f^{(m-1)}(a) = 0$ but $f^{(m)}(a) \neq 0$.

11.5 The Identity theorem

Theorem 11.7: Identity (Uniqueness) theorem

Let D be a domain and $f_1, f_2 \in \mathcal{H}(D)$. Suppose $f_1 = f_2$ on a set $S \subseteq D$ that has an accumulation point in D . Then $f_1 = f_2$ on all of D .

Proof. Let $g = f_1 - f_2 \in \mathcal{H}(D)$; then $g = 0$ on S , and we must show $g \equiv 0$ on D . Let

$$A = \left\{ z \in D : g^{(n)}(z) = 0 \text{ for all } n \geq 0 \right\}, \quad B = \left\{ z \in D : g^{(n)}(z) \neq 0 \text{ for some } n \geq 0 \right\}.$$

Clearly $D = A \sqcup B$.

B is open. If $z \in B$ then some $g^{(n)}(z) \neq 0$; by continuity $g^{(n)} \neq 0$ in a neighbourhood, so that neighbourhood is in B .

A is open. If $z_0 \in A$ then all derivatives vanish at z_0 , so the power series of g around z_0 is identically 0; hence $g \equiv 0$ on a disc $B(z_0, r) \subseteq D$, and on that disc all derivatives of g vanish, so $B(z_0, r) \subseteq A$.

A is nonempty. The accumulation point $a \in D$ of S lies in the zero set of g ; since zeros of a not-identically-zero holomorphic function are isolated, g cannot have a as an accumulation point of zeros *unless* $g \equiv 0$ near a . Hence g vanishes to infinite order at a , i.e. $a \in A$.

Now $D = A \sqcup B$ is a partition into two open sets with $A \neq \emptyset$. As D is connected, $B = \emptyset$ and $A = D$. Therefore $g \equiv 0$ on D , i.e. $f_1 = f_2$. ■

Corollary 11.2: $\mathcal{H}(D)$ is an integral domain

Let D be a domain and $f, g \in \mathcal{H}(D)$ with $f \cdot g \equiv 0$. Then $f \equiv 0$ or $g \equiv 0$.

Proof. Suppose $g \not\equiv 0$. Then the zeros of g are isolated, so $D \setminus g^{-1}(0)$ is a dense open subset of D on which $f = 0$ (because $fg \equiv 0$). This zero set of f has accumulation points throughout D ; by the Identity theorem $f \equiv 0$. ■

11.6 The Maximum-Modulus Principle

The proof rests on the **mean value property**: from the Cauchy Integral Formula with $z = a + re^{i\theta}$,

$$f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{i\theta}) d\theta.$$

Theorem 11.8: Maximum-Modulus Principle

Let D be a domain and $f \in \mathcal{H}(D)$.

- (1) **(Statement 1)** If $|f|$ attains a local maximum at some point $a \in D$, then f is constant on D .
- (2) **(Statement 2)** If D is bounded and f extends continuously to $\overline{D}$, then $|f|$ attains its maximum on the boundary ∂D .

Proof. (1) Suppose $|f(a)| \geq |f(z)|$ for z near a . By the mean value property, for small r ,

$$|f(a)| = \left| \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{i\theta}) d\theta \right| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})| d\theta \leq |f(a)|.$$

Equality throughout forces $|f(a + re^{i\theta})| = |f(a)|$ for all θ and all small r ; hence $|f|$ is constant on a neighbourhood of a . A holomorphic function of constant modulus on a domain is constant, so f is constant near a , and by the Identity theorem f is constant on D .

(2) $|f|$ is continuous on the compact set $\overline{D}$, so it attains a maximum at some $p \in \overline{D}$. If $p \in \partial D$ we are done. If $p \in D$, then by (1) f is constant on D , hence (by continuity) on $\overline{D}$, and the maximum is also attained on ∂D . ■

Corollary 11.3: Maximum on the boundary

For f holomorphic on a bounded domain D and continuous on $\overline{D}$,

$$\max_{z \in \overline{D}} |f(z)| = \max_{z \in \partial D} |f(z)|.$$

Theorem 11.9: Minimum-Modulus Principle

Let D be a domain and $f \in \mathcal{H}(D)$ with $f(z) \neq 0$ for all $z \in D$. If $|f|$ attains a local minimum in D , then f is constant. (Apply the maximum principle to $1/f$, which is holomorphic since f is non-vanishing.)

Remark

A **connected component** of an open set U is a maximal connected subset; open subsets of $\mathbb{C}$ decompose into countably many open connected components, and the theorems above are applied component by component.

11.7 The Open Mapping theorem

Theorem 11.10: Open Mapping theorem

Let D be a domain and $f \in \mathcal{H}(D)$ be non-constant. Then f is an **open map**: for every open $V \subseteq D$, the image $f(V)$ is open in $\mathbb{C}$.

Sketch. Fix $a \in D$ and let $w_0 = f(a)$. Since f is non-constant, $f(z) - w_0$ has a zero of some finite order $m \geq 1$ at a , and (as zeros are isolated) there is a small circle $|z - a| = r$ on which $|f(z) - w_0| \geq \delta > 0$. For any w with $|w - w_0| < \delta$, Rouché-type reasoning (or the argument principle) shows $f(z) - w$ has the same number m of zeros inside the circle; in particular f takes the value w . Hence $B(w_0, \delta) \subseteq f(V)$, so $f(V)$ is open. ■

11.8 The Inverse Function theorem

Theorem 11.11: Inverse Function theorem (real form)

Let $F : U(\subseteq \mathbb{R}^n) \rightarrow \mathbb{R}^n$ be C^1 and suppose the Jacobian $DF(a)$ is invertible at a . Then F is a local diffeomorphism near a : there are neighbourhoods of a and $F(a)$ on which F is a bijection with C^1 inverse.

Theorem 11.12: Inverse Function theorem (holomorphic form)

Let $f \in \mathcal{H}(U)$ and suppose $f'(a) \neq 0$. Then there is a neighbourhood V of a such that $f|_V$ is injective, $f(V)$ is open, and the inverse $f^{-1} : f(V) \rightarrow V$ is holomorphic with

$$(f^{-1})'(w) = \frac{1}{f'(f^{-1}(w))}.$$

Sketch. $f'(a) \neq 0$ means the real Jacobian (which is the multiplication matrix of $f'(a)$, with determinant $|f'(a)|^2$) is invertible; the real inverse function theorem gives a local C^1 inverse g . The formula $g'(w) = 1/f'(g(w))$ from the holomorphic-inverse theorem shows g is holomorphic. ■

11.8.1 Non-vanishing of the derivative

Theorem 11.13: Local k -th root and injectivity

Let $f \in \mathcal{H}(U)$ and $a \in U$ with $f'(a) \neq 0$. Then near a one can write, after a holomorphic change of coordinates, $f(z) - f(a) = (h(z))^k$ with h holomorphic and $h'(a) \neq 0$; more generally, if $f(z) - f(a)$ vanishes to order k at a then $f(z) - f(a) = (h(z))^k$ locally, where

$$h(z) = \exp\left(\frac{1}{k} \log g(z)\right), \quad f(z) - f(a) = (z - a)^k g(z), \quad g(a) \neq 0,$$

choosing a branch of $\log$ near $g(a) \neq 0$. Consequently f is locally k -to-one, and 1-to-one exactly when $f'(a) \neq 0$ (i.e. $k = 1$).

Sketch. Write $f(z) - f(a) = (z-a)^k g(z)$ with $g(a) \neq 0$, g holomorphic. Since $g(a) \neq 0$, a branch of $\log g$ exists near a ; set $h(z) = (z-a) \exp(\frac{1}{k} \log g(z))$. Then $h(z)^k = (z-a)^k g(z) = f(z) - f(a)$ and $h'(a) \neq 0$, so h is a local biholomorphism and f behaves like $w \mapsto w^k$ in the h -coordinate — k -to-one for $k \geq 2$, injective for $k = 1$. ■

CHAPTER 12

Isolated Singularities

12.1 Isolated singularities

Definition 12.1: Isolated singularity

Let f be holomorphic on a deleted neighbourhood $B(\alpha, \delta) \setminus \{\alpha\}$ of a point α (but not necessarily at α itself). Then α is called an **isolated singularity** of f .

Example 12.1: Isolated vs non-isolated singularities

- (a) $f(z) = \frac{1}{z}$ has an isolated singularity at 0.
- (b) A branch of $\log z$ is not holomorphic on any deleted neighbourhood of 0 (the whole slit ray is singular), so 0 is *not* an isolated singularity of $\log$.
- (c) $\tan\left(\frac{1}{z}\right)$ has singularities at the points where $\frac{1}{z} = \frac{\pi}{2} + n\pi$, i.e. $z = \frac{1}{\frac{\pi}{2} + n\pi}$; these accumulate at 0, so 0 is a *non-isolated* singularity, while each nonzero such z is isolated.

Definition 12.2: The three types

An isolated singularity α of f is called:

- (1) a **removable singularity** if f can be extended holomorphically across α (there is a holomorphic g on $B(\alpha, \delta)$ with $g = f$ on the deleted neighbourhood);
- (2) a **polar singularity** (a **pole**) if on a deleted neighbourhood $B(\alpha, \delta) \setminus \{\alpha\}$ we can write

$$f(z) = \frac{g(z)}{h(z)}, \quad g(\alpha) \neq 0, \quad h(\alpha) = 0, \quad g, h \in \mathcal{H}(B(\alpha, \delta));$$

(e.g. $f(z) = 1/z$ — the limit would not exist);

- (3) an **essential singularity** if α is neither a removable nor a polar singularity (e.g. $\exp(1/z)$ at $z = 0$).

12.2 Characterisation of removable singularities

Theorem 12.1: Removable singularity

Let α be an isolated singularity of f . The following are equivalent:

$$\alpha \text{ is removable} \iff \lim_{z \rightarrow \alpha} f(z) \text{ exists} \iff \lim_{z \rightarrow \alpha} (z - \alpha)f(z) = 0.$$

Proof. $(\Rightarrow)$ is immediate: if f extends holomorphically then $\lim_{z \rightarrow \alpha} f(z)$ exists (equal to the

extended value) and $\lim(z - \alpha)f(z) = 0$.

($\Leftarrow$) Suppose $\lim_{z \rightarrow \alpha}(z - \alpha)f(z) = 0$. Define

$$h(z) = \begin{cases} (z - \alpha)f(z), & z \neq \alpha, \\ 0, & z = \alpha, \end{cases}$$

which is continuous at α (by hypothesis) and holomorphic on the deleted neighbourhood, hence holomorphic on $B(\alpha, \delta)$. Now define

$$g(z) = \begin{cases} \frac{h(z) - h(\alpha)}{z - \alpha}, & z \neq \alpha, \\ h'(\alpha), & z = \alpha, \end{cases}$$

which is again holomorphic around α . For $z \neq \alpha$,

$$g(z) = \frac{h(z) - 0}{z - \alpha} = \frac{(z - \alpha)f(z)}{z - \alpha} = f(z),$$

so g is the required holomorphic extension of f ; thus α is a removable singularity. $\blacksquare$

12.3 Characterisation of poles

Theorem 12.2: Pole

Let α be an isolated singularity of f . Then

$$\alpha \text{ is a pole} \iff \lim_{z \rightarrow \alpha} |f(z)| = \infty.$$

Moreover, near a pole one can write $f(z) = \frac{g(z)}{(z - \alpha)^m}$ with g holomorphic and $g(\alpha) \neq 0$; the integer $m \geq 1$ is the **order** of the pole.

Proof. ($\Rightarrow$) Immediate: if $f = g/h$ with $g(\alpha) \neq 0$, $h(\alpha) = 0$, then $|f(z)| \rightarrow \infty$.

($\Leftarrow$) Suppose $\lim_{z \rightarrow \alpha} |f(z)| = \infty$. Then there is $\delta > 0$ with $|f(z)| > 1$ (indeed $> 10^{2026}$, say) on $B(\alpha, \delta) \setminus \{\alpha\}$. Define

$$h(z) = \frac{1}{f(z)}, \quad z \in B(\alpha, \delta) \setminus \{\alpha\},$$

which is holomorphic there, and $\lim_{z \rightarrow \alpha} h(z) = 0$. So h has a removable singularity at α and extends with $h(\alpha) = 0$. Write its power series

$$h(z) = \sum_{n \geq m} a_n (z - \alpha)^n, \quad a_m \neq 0, \quad a_i = 0 \quad (0 \leq i < m),$$

so $h(z) = (z - \alpha)^m \underbrace{\sum_{n \geq m} a_n (z - \alpha)^{n-m}}_{=: h_1(z)}$, with h_1 holomorphic on $B(\alpha, \delta)$ and $h_1(\alpha) = a_m \neq 0$.

Hence there is $\delta' > 0$ with $h_1 \neq 0$ on $B(\alpha, \delta')$, so $1/h_1$ is holomorphic there. Then on $B(\alpha, \delta') \setminus \{\alpha\}$,

$$f(z) = \frac{1}{h(z)} = \frac{1}{(z - \alpha)^m h_1(z)} = \frac{1/h_1(z)}{(z - \alpha)^m},$$

i.e. $f(z) = \frac{g(z)}{(z - \alpha)^m}$ with $g = 1/h_1$ holomorphic and $g(\alpha) \neq 0$. Thus α is a pole of order m . $\blacksquare$

Remark

The proof shows: if α is a pole of f , then around α we may write $f(z) = \frac{g(z)}{(z - \alpha)^m}$ for some $m > 0$ with $g(\alpha) \neq 0$ and g holomorphic at α . We say α is a pole of “**order** m ”. Also: α is a pole of order $k \iff (z - \alpha)^k f(z)$ has a removable singularity at α .

12.4 Characterisation of essential singularities

Theorem 12.3: Essential singularity

Let α be an isolated singularity of f . Then α is an essential singularity if and only if there exist two sequences $a_n \rightarrow \alpha$ (with $a_n \neq \alpha$) and $b_n \rightarrow \alpha$ (with $b_n \neq \alpha$) such that

$$f(a_n) \text{ is bounded,} \quad |f(b_n)| \rightarrow \infty \quad (n \rightarrow \infty).$$

Equivalently, there are sequences $a_n, b_n \rightarrow \alpha$ with $f(a_n) \rightarrow c$ and $f(b_n) \rightarrow d$ for some $c \neq d$.

Example 12.2: $\exp(1/z)$ has an essential singularity at 0

Take $z_n = \frac{1}{n} \rightarrow 0$: then $\exp(1/z_n) = e^n \rightarrow \infty$. Take $z_n = \frac{i}{n} \rightarrow 0$: then $\exp(1/z_n) = e^{-in}$, which stays bounded (modulus 1). The two behaviours differ, so 0 is an essential singularity of $\exp(1/z)$.

Remark

Exercise. $\sin(\frac{1}{z})$ and $\cos(\frac{1}{z})$ have essential singularities at 0.

12.5 Casorati–Weierstrass and Picard

Theorem 12.4: Casorati–Weierstrass

Let α be an essential singularity of f . Then for any $\delta > 0$, the image

$$f(B(\alpha, \delta) \setminus \{\alpha\}) \text{ is dense in } \mathbb{C}.$$

Proof. It suffices to show $f(B(\alpha, \delta) \setminus \{\alpha\})$ meets every disc $B(a, s)$. Suppose not: there exist $a \in \mathbb{C}$ and $s > 0$ with

$$f(B(\alpha, \delta) \setminus \{\alpha\}) \cap B(a, s) = \emptyset,$$

i.e. $|f(z) - a| \geq s > 0$ on $B(\alpha, \delta) \setminus \{\alpha\}$. Define

$$g(z) = \frac{1}{f(z) - a}, \quad z \in B(\alpha, \delta) \setminus \{\alpha\},$$

holomorphic on the deleted neighbourhood with $|g(z)| \leq \frac{1}{s}$ (bounded). Hence g has a *removable* singularity at α . On the deleted neighbourhood,

$$f(z) - a = \frac{1}{g(z)}.$$

If $g(\alpha) \neq 0$, then $f - a$ (hence f) extends holomorphically: α is a removable singularity of f . If $g(\alpha) = 0$, then $f - a$ has a pole at α , so α is a polar singularity of f . Either way α is not essential — a **contradiction**. Therefore

$$f(B(\alpha, \delta) \setminus \{\alpha\}) \cap B(a, s) \neq \emptyset,$$

and density follows. ■

Theorem 12.5: Great Picard's theorem (statement)

A much stronger statement holds than Casorati–Weierstrass: near an essential singularity α ,

$$f(B(\alpha, \delta) \setminus \{\alpha\}) = \mathbb{C} \quad \text{or} \quad \mathbb{C} \setminus \{a\}$$

for some single exceptional value a . That is, on every deleted neighbourhood f takes every complex value, with at most one exception, infinitely often.

CHAPTER 13

Laurent Series and Residues

Around a removable singularity a function has an ordinary power series; around a pole or an essential singularity we need *negative* powers too. This leads to Laurent series.

13.1 Double-sided series

Definition 13.1: Convergence of a double-sided series

Let $(f_n)_{n \in \mathbb{Z}}$ be a doubly-infinite sequence of functions. We say $\sum_{n \in \mathbb{Z}} f_n$ **converges** (absolutely / normally / uniformly) if both

$$\sum_{n \geq 0} f_n \quad \text{and} \quad \sum_{n < 0} f_n$$

converge (absolutely / normally / uniformly).

Definition 13.2: Formal Laurent series

A **formal Laurent series** is $\sum_{n \in \mathbb{Z}} a_n x^n$. Split it as

$$\sum_{n \in \mathbb{Z}} a_n x^n = \underbrace{\sum_{n \geq 0} a_n x^n}_{\text{ROC} = \rho_1} + \underbrace{\sum_{n < 0} a_n x^n}_{\text{via } \sum_{n < 0} a_n (1/x)^{-n}, \text{ ROC} = \frac{1}{\rho_2}}.$$

13.1.1 Region of convergence: an annulus

The “positive part” $\sum_{n \geq 0} a_n z^n$ converges for $|z| < \rho_1$ (uniformly on compact subsets). The “negative part” $\sum_{n < 0} a_n z^n = \sum_{n < 0} a_n (1/z)^{-n}$ converges for $|1/z| < \frac{1}{\rho_2}$, i.e. $\rho_2 < |z|$. Hence the full series

$$\sum_{n \in \mathbb{Z}} a_n z^n \quad \text{converges for} \quad \rho_2 < |z| < \rho_1,$$

an **annulus**. Shifting the centre, $\sum_{n \in \mathbb{Z}} a_n (z - \alpha)^n$ defines a holomorphic function on the annulus $\rho_2 < |z - \alpha| < \rho_1$, where ρ_1 is the ROC of $\sum_{n \geq 0} a_n x^n$ and $\frac{1}{\rho_2}$ is the ROC of $\sum_{n < 0} a_n x^{-n}$.

Remark

Recall a power series defines a holomorphic function on a *disc*; a Laurent series defines a holomorphic function on an *annulus*. If we prove that any holomorphic function on an annulus has a Laurent expansion, then for any isolated singularity α of f , taking the annulus $0 < |z - \alpha| < \delta$, f has a Laurent expansion around α .

13.2 Existence of the Laurent expansion

Theorem 13.1: Laurent expansion on an annulus

A holomorphic function on an annulus $\rho_2 < |z - \alpha| < \rho_1$ has a Laurent expansion valid on the annulus. Explicitly, choosing radii $\rho_2 < R_2 < |\alpha - a| < R_1 < \rho_1$, for α in the annulus

$$f(\alpha) = \frac{1}{2\pi i} \int_{a+\gamma_{R_1}} \frac{f(z)}{z - \alpha} dz - \frac{1}{2\pi i} \int_{a+\gamma_{R_2}} \frac{f(z)}{z - \alpha} dz,$$

and consequently

$$f(\alpha) = \sum_{n \geq 0} c_n (\alpha - a)^n + \sum_{n < 0} d_n (\alpha - a)^n,$$

where

$$c_n = \frac{1}{2\pi i} \int_{a+\gamma_{R_1}} \frac{f(z)}{(z - a)^{n+1}} dz, \quad d_n = \frac{1}{2\pi i} \int_{a+\gamma_{R_2}} \frac{f(z)}{(z - a)^{n+1}} dz.$$

Proof. Step 1: the two-circle formula. Define, as before,

$$g(z) = \begin{cases} \frac{f(z) - f(\alpha)}{z - \alpha}, & z \neq \alpha, \\ f'(\alpha), & z = \alpha, \end{cases}$$

holomorphic on the annulus. We claim

$$\int_{a+\gamma_{R_1}} g(z) dz = \int_{a+\gamma_{R_2}} g(z) dz.$$

Divide the annular region between the two circles into finitely many pieces using radial *spokes*. Cut so that each piece fits inside a disc contained in the annulus. Each such piece is bounded by a closed curve γ' lying in a disc, on which g is holomorphic; by the closed-curve theorem $\int_{\gamma'} g dz = 0$. Summing over all pieces, the spoke edges are each traversed twice in opposite directions and cancel, leaving the outer circle (anticlockwise) minus the inner circle (which appears clockwise, hence with a minus sign):

$$\int_{a+\gamma_{R_1}} g dz - \int_{a+\gamma_{R_2}} g dz = 0.$$

Writing out g and using $\int_{a+\gamma_{R_1}} \frac{dz}{z - \alpha} = 2\pi i$ while $\int_{a+\gamma_{R_2}} \frac{dz}{z - \alpha} = 0$ (as α lies outside the inner circle, where $\frac{1}{z - \alpha}$ has a primitive) gives

$$f(\alpha) = \frac{1}{2\pi i} \int_{a+\gamma_{R_1}} \frac{f(z)}{z - \alpha} dz - \frac{1}{2\pi i} \int_{a+\gamma_{R_2}} \frac{f(z)}{z - \alpha} dz.$$

Step 2: expand each kernel. On the outer circle $a + \gamma_{R_1}$ we have $|\frac{\alpha - a}{z - a}| < 1$, so

$$\frac{1}{z - \alpha} = \frac{1}{(z - a) - (\alpha - a)} = \frac{1}{z - a} \sum_{n \geq 0} \left(\frac{\alpha - a}{z - a} \right)^n = \sum_{n \geq 0} \frac{(\alpha - a)^n}{(z - a)^{n+1}}.$$

On the inner circle $a + \gamma_{R_2}$ we have $|a - z| < |a - \alpha|$, so

$$\frac{1}{z - \alpha} = \frac{1}{(a - \alpha) - (a - z)} = \frac{1}{a - \alpha} \cdot \frac{1}{1 - \frac{a - z}{a - \alpha}} = - \sum_{n \geq 0} \frac{(z - a)^n}{(\alpha - a)^{n+1}}.$$

Substituting these uniformly convergent expansions and integrating term by term,

$$f(\alpha) = \sum_{n \geq 0} \underbrace{\left(\frac{1}{2\pi i} \int_{\alpha + \gamma_{R_1}} \frac{f(z)}{(z - \alpha)^{n+1}} dz \right)}_{c_n} (\alpha - \alpha)^n + \sum_{n < 0} \underbrace{\left(\frac{1}{2\pi i} \int_{\alpha + \gamma_{R_2}} \frac{f(z)}{(z - \alpha)^{n+1}} dz \right)}_{d_n} (\alpha - \alpha)^n.$$

This is the Laurent expansion. ■

Corollary 13.1: Laurent expansion at an isolated singularity

Around a point α of isolated singularity, a function has a Laurent expansion $f(z) = \sum_{n \in \mathbb{Z}} a_n (z - \alpha)^n$ (valid on $0 < |z - \alpha| < \delta$). The part $\sum_{n < 0} a_n (z - \alpha)^n$ is called the **principal part** of the Laurent series.

13.3 Classification via the principal part

Removable singularity	$\iff$	principal part = 0
Polar singularity (pole)	$\iff$	principal part is a <i>finite</i> sum
Essential singularity	$\iff$	principal part is an <i>infinite</i> sum

13.4 The residue

Theorem 13.2: Integral theorem (residue form)

Let α be an isolated singularity of f , with f holomorphic on $B(\alpha, \delta) \setminus \{\alpha\}$. For $0 < R < \delta$,

$$\int_{\alpha + \gamma_R} f(z) dz = 2\pi i c_{-1},$$

where c_{-1} is the coefficient of $(z - \alpha)^{-1}$ in the Laurent expansion of f around α .

Proof. Write $f(z) = \sum_{n \in \mathbb{Z}} a_n (z - \alpha)^n$, converging uniformly on the circle $|z - \alpha| = R$. Integrating term by term and using the fundamental example

$$\int_{\alpha + \gamma_R} (z - \alpha)^n dz = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1, \end{cases}$$

only the $n = -1$ term survives:

$$\int_{\alpha + \gamma_R} f(z) dz = a_{-1} \cdot 2\pi i = 2\pi i c_{-1}. \quad \blacksquare$$

Definition 13.3: Residue

The coefficient of $(z - \alpha)^{-1}$ in the Laurent expansion of f around α is called the **residue** of f at α , written $\text{Res}(f, \alpha)$. Thus

$$\int_{\alpha + \gamma_R} f(z) dz = 2\pi i \text{Res}(f, \alpha).$$

13.5 Calculating residues

13.5.1 Removable singularity

If α is removable there are no negative powers, so

$$\text{Res}(f, \alpha) = 0.$$

13.5.2 Simple pole (order 1)

Let α be a simple pole. Then $f(z) = \sum_{n \geq -1} c_n (z - \alpha)^n$, so

$$(z - \alpha)f(z) = \sum_{n \geq -1} c_n (z - \alpha)^{n+1} = c_{-1} + c_0(z - \alpha) + c_1(z - \alpha)^2 + \cdots,$$

a convergent power series (same ROC). Evaluating at $z = \alpha$,

$$\text{Res}(f, \alpha) = c_{-1} = \lim_{z \rightarrow \alpha} (z - \alpha) f(z) \quad (\text{simple pole}).$$

13.5.3 Quotient form $f = A/B$ with a simple zero of B

Suppose $f(z) = \frac{A(z)}{B(z)}$ with A, B holomorphic, $A(\alpha) \neq 0$, $B(\alpha) = 0$ and $B'(\alpha) \neq 0$ (a simple zero of B , hence a simple pole of f). Then

$$\lim_{z \rightarrow \alpha} (z - \alpha)f(z) = \lim_{z \rightarrow \alpha} (z - \alpha) \frac{A(z)}{B(z)} = \lim_{z \rightarrow \alpha} \frac{A(z)}{\frac{B(z) - B(\alpha)}{z - \alpha}} = \frac{A(\alpha)}{B'(\alpha)},$$

so

$$\text{Res}(f, \alpha) = c_{-1} = \frac{A(\alpha)}{B'(\alpha)} \quad (\text{simple zero of the denominator}).$$

13.5.4 Pole of order m

Theorem 13.3: Residue at a pole of order m

Let α be a pole of order $m \geq 1$ of f , and set $g(z) = (z - \alpha)^m f(z)$ (holomorphic at α). Then

$$\operatorname{Res}(f, \alpha) = \frac{1}{(m-1)!} \lim_{z \rightarrow \alpha} g^{(m-1)}(z).$$

Idea. Near α , $f(z) = \sum_{n \geq -m} c_n (z - \alpha)^n$, so $g(z) = (z - \alpha)^m f(z) = \sum_{n \geq -m} c_n (z - \alpha)^{n+m} = \sum_{k \geq 0} c_{k-m} (z - \alpha)^k$ is holomorphic, with the coefficient of $(z - \alpha)^{m-1}$ equal to c_{-1} . By the power-series formula for coefficients, $c_{-1} = \frac{g^{(m-1)}(\alpha)}{(m-1)!}$. ■

13.5.5 Essential singularity**! Caution**

Unfortunately, there is **no explicit closed-form formula** for calculating the residue at an essential singularity. One must compute the Laurent expansion directly and read off the coefficient c_{-1} .

End of course.